Homework Assignments 1

Chapter 4 Decision Making for a Single Sample

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報告概述 / Overview

這次作業總共 10 題，全部出自 Montgomery 教科書第 4 章 *Decision Making for a Single Sample*。10 題依題型大致可以分成五群：

- 已知 σ 的平均數推論：4-42、4-44，用常態的 z 檢定。
- 未知 σ 的平均數推論：4-54、4-59、4-65，用 t 檢定，順便檢查資料是否近似常態。
- 單一母體標準差：4-71，用 χ^2 檢定。
- 單一比例：4-75、4-76、4-89，用 z 近似，最後一題加碼比較傳統 Wald CI 跟 Agresti-Coull 兩種寫法。
- 預測與容忍區間：4-95，要把 PI 跟 TI 區分清楚。

每題的處理流程都跟教科書範本一致：先把資料 hard-code 進來、印描述統計、做必要的常態性檢查、跑檢定算 P-value，必要時補上 power、樣本數，再加幾張視覺化的圖。所有數值依 AGENTS.md 的慣例四捨五入到小數第四位（樣本數一律 `ceiling()`）。

整份報告的閱讀順序是：題目原文 → R 程式 → Console 輸出 → 圖 → 最終答案。共用工具函式（如 `Pwr.z.test`、`Pwr.t.test.custom`、`find_min_n_t`、`Pwr.chisq.test`、`prop_wald_ci`、`agresti_coull_ci`、`prediction_interval_normal`）跟套件載入都統一放在腳本最前段，完整 R 原始碼在文末的「附錄 A」一次列出。

Exercise 4-42: Thermocouple Life

> **4-42.** The life in hours of a thermocouple used in a furnace is known to be approximately normally distributed, with standard deviation $\sigma = 20$ hours. A random sample of 15 thermocouples resulted in the following data:
> 553, 552, 567, 579, 550, 541, 537, 553, 552, 546, 538, 553, 581, 539, 529.
> (a) Is there evidence to support the claim that mean life exceeds 540 hours? Use a fixed-level test with $\alpha = 0.05$.
> (b) What is the P-value for this test?
> (c) What is the β-value for this test if the true mean life is 560 hours?
> (d) What sample size would be required to ensure that β does not exceed 0.10 if the true mean life is 560 hours?
> (e) Construct a 95% one-sided lower CI on the mean life.
> (f) Use the CI found in part (e) to test the hypothesis.

```r
# ==============================================================================
# EXERCISE 4-42: Thermocouple Life (Section 4-4)
# 已知 sigma=20, n=15, mu0=540, alpha=0.05, H1: mu > 540
# ==============================================================================
print_header("EXERCISE 4-42: THERMOCOUPLE LIFE")

# 原始資料，照題目 hard-code，不可改
thermo <- c(553, 552, 567, 579, 550, 541, 537, 553, 552,
            546, 538, 553, 581, 539, 529)

params_42 <- list(mu0 = 540, sigma = 20, alpha = 0.05, n = length(thermo))

print_subhead("Summary statistics")
print(get_stats(thermo))
cat(sprintf("Known sigma = %.4f\n", params_42$sigma))
cat(sprintf("Known SE    = %.4f\n", params_42$sigma / sqrt(params_42$n)))

# (a) z 檢定，one-sided greater
print_subhead("(a)(b) One-sided z-test, P-value approach")
(xbar     <- mean(thermo))
(se_42    <- params_42$sigma / sqrt(params_42$n))
(z0_42    <- (xbar - params_42$mu0) / se_42)
(p_42     <- 1 - pnorm(z0_42))
crit_42   <- qnorm(1 - params_42$alpha)

cat("H0: mu <= 540\n")
cat("H1: mu >  540\n")
cat(sprintf("alpha       = %.2f\n", params_42$alpha))
cat(sprintf("xbar        = %.4f\n", xbar))
cat(sprintf("z0          = %.4f\n", z0_42))
cat(sprintf("z critical  = %.4f\n", crit_42))
cat(sprintf("P-value     = %.4f\n", p_42))
cat("Decision: ",
    if (p_42 < params_42$alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")
cat("Conclusion: ",
    if (p_42 < params_42$alpha)
      "There is evidence that mean life exceeds 540 hours.\n"
    else
      "Insufficient evidence that mean life exceeds 540 hours.\n")
```

```r
# 套件交叉驗證
zchk_42 <- BSDA::z.test(thermo, mu = params_42$mu0,
                        sigma.x = params_42$sigma,
                        alternative = "greater")
cat(sprintf("[Cross-check BSDA::z.test]  z = %.4f, p = %.4f\n",
            zchk_42$statistic, zchk_42$p.value))

# (c) Beta，當真實 mu = 560
print_subhead("(c) Beta when true mu = 560")
# 拒絕門檻換算成 xbar 比較直觀
xbar_crit_42 <- params_42$mu0 + crit_42 * se_42
mu_true_42   <- 560
(beta_42c    <- pnorm((xbar_crit_42 - mu_true_42) / se_42))
power_42c    <- 1 - beta_42c
cat(sprintf("xbar critical = %.4f\n", xbar_crit_42))
cat(sprintf("beta          = %.4f\n", beta_42c))
cat(sprintf("power         = %.4f\n", power_42c))

# (d) 要 beta <= 0.10 需要的 n
print_subhead("(d) Sample size so beta <= 0.10 when true mu = 560")
delta_42d <- mu_true_42 - params_42$mu0
n_req_42d <- SplSz.z.test(delta = delta_42d, sigma = params_42$sigma,
                          alpha = params_42$alpha, pwr = 0.90, side = 1)
pw_minus  <- Pwr.z.test(n_req_42d - 1, delta_42d, params_42$sigma,
                        params_42$alpha, alt = "greater")
pw_at     <- Pwr.z.test(n_req_42d,     delta_42d, params_42$sigma,
                        params_42$alpha, alt = "greater")
cat(sprintf("Required n         = %d\n", n_req_42d))
cat(sprintf("Power at n-1 (%2d) = %.4f , beta = %.4f\n",
            n_req_42d - 1, pw_minus, 1 - pw_minus))
cat(sprintf("Power at n   (%2d) = %.4f , beta = %.4f\n",
            n_req_42d,     pw_at,    1 - pw_at))

# (e) 95% one-sided lower CI on mu
print_subhead("(e) 95% one-sided lower CI on mean life")
LCL_42 <- xbar - crit_42 * se_42
cat(sprintf("LCL = %.4f\n", LCL_42))
cat(sprintf("CI : [ %.4f , Inf )\n", LCL_42))

# (f) 用 CI 做檢定
print_subhead("(f) Test the hypothesis using the CI in (e)")
cat(sprintf("LCL = %.4f, mu0 = %d => ", LCL_42, params_42$mu0))
if (LCL_42 > params_42$mu0) {
  cat("LCL > 540, the null value lies outside the CI, reject H0.\n")
} else {
  cat("LCL <= 540, the null value lies inside the CI, do not reject H0.\n")
}

# Power curve 視覺化
power_grid_42 <- tibble(
  mu_true = seq(540, 580, length.out = 200)
) %>% mutate(
  power = Pwr.z.test(params_42$n, mu_true - params_42$mu0,
```

```
                    params_42$sigma, params_42$alpha, alt = "greater")
)
p_pow_42 <- ggplot(power_grid_42, aes(mu_true, power)) +
  geom_line(color = "steelblue", linewidth = 1) +
  geom_hline(yintercept = 0.90, linetype = "dashed", color = "firebrick") +
  geom_vline(xintercept = 560, linetype = "dotted",  color = "darkgreen") +
  labs(title = "Ex 4-42 Power curve",
       subtitle = "known   = 20, n = 15,   = 0.05",
       x = "True mean life", y = "Power") +
  theme_minimal(base_size = 12)
ggsave("plots/ex4_42_power_curve.png", p_pow_42,
       width = 6.5, height = 4.5, dpi = 200)
#>
#> ====================================================
#>  EXERCISE 4-42: THERMOCOUPLE LIFE
#> ====================================================
#>
#> --- Summary statistics ---
#>    n     Mean     Var Std_Dev Std_Error Median
#> 1 15 551.3333 219.381 14.8115    3.8243    552
#> Known sigma = 20.0000
#> Known SE    = 5.1640
#>
#> --- (a)(b) One-sided z-test, P-value approach ---
#> [1] 551.3333
#> [1] 5.163978
#> [1] 2.194691
#> [1] 0.0140929
#> H0: mu <= 540
#> H1: mu >  540
#> alpha       = 0.05
#> xbar        = 551.3333
#> z0          = 2.1947
#> z critical  = 1.6449
#> P-value     = 0.0141
#> Decision:  REJECT H0
#> Conclusion:  There is evidence that mean life exceeds 540 hours.
#> [Cross-check BSDA::z.test]  z = 2.1947, p = 0.0141
#>
#> --- (c) Beta when true mu = 560 ---
#> [1] 0.01293593
#> xbar critical = 548.4940
#> beta          = 0.0129
#> power         = 0.9871
#>
#> --- (d) Sample size so beta <= 0.10 when true mu = 560 ---
#> Required n          = 9
#> Power at n-1 ( 8) = 0.8817 , beta = 0.1183
#> Power at n   ( 9) = 0.9123 , beta = 0.0877
#>
#> --- (e) 95% one-sided lower CI on mean life ---
#> LCL = 542.8393
#> CI : [ 542.8393 , Inf )
#>
```

```
#> --- (f) Test the hypothesis using the CI in (e) ---
#> LCL = 542.8393, mu0 = 540 => LCL > 540, the null value lies outside the CI, reject H0.
```

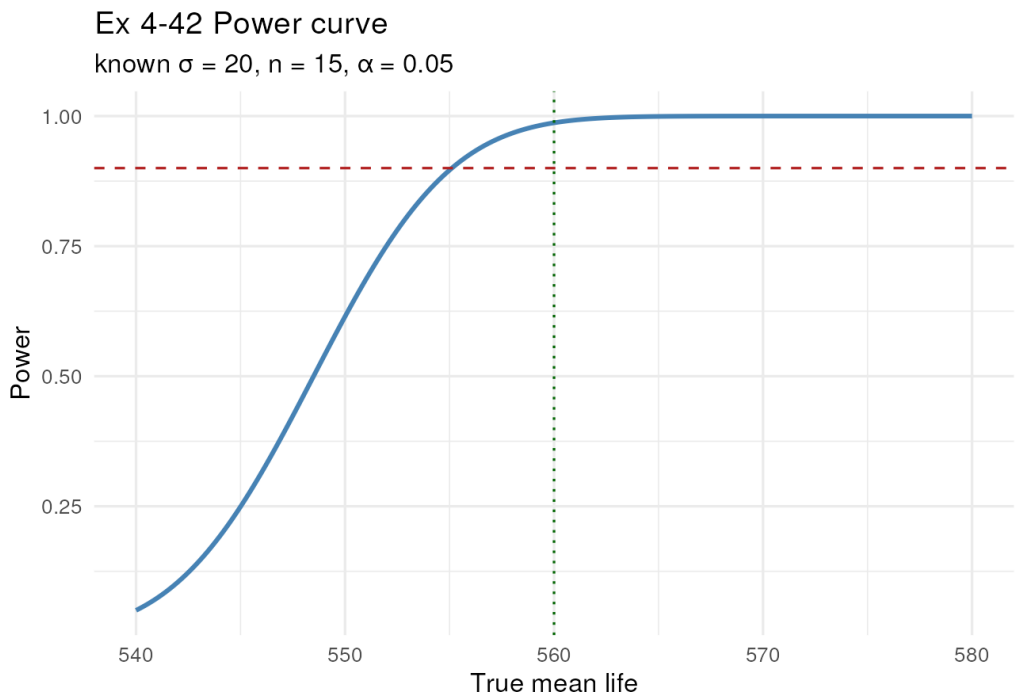

Figure 1: Ex 4-42 Power curve（紅虛線是目標 power $= 0.9$，綠點線是 $\mu_{true} = 560$）

最終答案 Final Answer

- **(a)(b)** 算出 $z_0 = 2.1947$、P-value $= 0.0141$，比 $\alpha = 0.05$ 小，所以 **REJECT** H_0；資料站得住腳，平均壽命確實超過 540 小時。
- **(c)** 假設真實平均跑到 $\mu = 560$，$\beta \approx 0.0129$、power ≈ 0.9871，這個 $n = 15$ 已經很敏感了。
- **(d)** 要把 β 壓到 0.10 以下：$n = 8$ 時 power 只有 0.8817 還差一點，$n = 9$ 時 power 衝到 0.9123 過關，所以 $n = \mathbf{9}$。
- **(e)** 95% 一邊下界 CI：$[542.8393,\ \infty)$。
- **(f)** 下界 542.8393 已經跑到 540 上面去了，0 假設值掉到 CI 之外，自然也是 **REJECT** H_0，跟 (a) 結論一致。

Exercise 4-44: Sample Size for Estimating the Mean

> **4-44.** Suppose that in Exercise 4-42 we wanted to be 95% confident that the error in estimating the mean life is less than 5 hours. What sample size should we use?

```r
# ==============================================================================
# EXERCISE 4-44: Sample size for estimating the mean (Section 4-4)
# 沿用 4-42 的 sigma = 20，要 95% 信賴、誤差 < 5 小時
# ==============================================================================
print_header("EXERCISE 4-44: SAMPLE SIZE FOR MEAN LIFE")

E_44       <- 5
sigma_44   <- 20
conf_44    <- 0.95
n_44       <- SplSz.z.CI(E = E_44, sigma = sigma_44,
                         conf_level = conf_44, side = 2)

cat(sprintf("Confidence level   = %.2f\n", conf_44))
cat(sprintf("Margin of error E  = %.0f\n", E_44))
cat(sprintf("sigma              = %.0f\n", sigma_44))
cat(sprintf("Required sample n  = %d\n", n_44))
# 用公式手算對一下
z_44 <- qnorm(1 - (1 - conf_44)/2)
cat(sprintf("Manual: ((z * sigma)/E)^2 = ((%.4f * %d)/%d)^2 = %.4f -> ceiling = %d\n",
            z_44, sigma_44, E_44, (z_44 * sigma_44 / E_44)^2, n_44))
#>
#> ====================================================
#>  EXERCISE 4-44: SAMPLE SIZE FOR MEAN LIFE
#> ====================================================
#> Confidence level   = 0.95
#> Margin of error E  = 5
#> sigma              = 20
#> Required sample n  = 62
#> Manual: ((z * sigma)/E)^2 = ((1.9600 * 20)/5)^2 = 61.4633 -> ceiling = 62
```

最終答案 Final Answer

直接套已知 σ 的 margin-of-error 公式：

$$n = \left\lceil \left(\frac{z_{0.025}\,\sigma}{E} \right)^2 \right\rceil = \left\lceil \left(\frac{1.96 \times 20}{5} \right)^2 \right\rceil = \lceil 61.4633 \rceil = \mathbf{62}\,.$$

Exercise 4-54: Interior Temperature

> **4-54.** An article in the *ASCE Journal of Energy Engineering* (Vol. 125, 1999, pp. 59-75) describes a study of the thermal inertia properties of autoclaved aerated concrete used as a building material. Five samples of the material were tested in a structure, and the average interior temperature (°C) reported was as follows:
> 23.01, 22.22, 22.04, 22.62, and 22.59.
> (a) Test the hypotheses $H_0 : \mu = 22.5$ versus $H_1 : \mu \neq 22.5$, using $\alpha = 0.05$. Use the P-value approach.
> (b) Check the assumption that interior temperature is normally distributed.
> (c) Find a 95% CI on the mean interior temperature.
> (d) What sample size would be required to detect a true mean interior temperature as high as 22.75 if we wanted the power of the test to be at least 0.9? Use the sample standard deviation s as an estimate of σ.

```r
# ==============================================================================
# EXERCISE 4-54: Interior Temperature  (Section 4-5)
# 未知 sigma, n=5, mu0=22.5, alpha=0.05, H1: mu != 22.5
# ==============================================================================
print_header("EXERCISE 4-54: INTERIOR TEMPERATURE")

temp_int <- c(23.01, 22.22, 22.04, 22.62, 22.59)
params_54 <- list(mu0 = 22.5, alpha = 0.05, n = length(temp_int))

print_subhead("Summary statistics")
print(get_stats(temp_int))

# (b) 常態性檢查
print_subhead("(b) Normality check  (Shapiro-Wilk + Q-Q plot)")
(sw_54 <- shapiro.test(temp_int))
cat(" 樣本數很小，Shapiro-Wilk power 不高，需要搭配 Q-Q plot 一起看。\n")
save_normality_plots(temp_int, "Ex 4-54 Interior Temperature", "ex4_54")
cat("Plots saved: plots/ex4_54_qq.png, plots/ex4_54_hist.png\n")

# (a) t 檢定，雙尾
print_subhead("(a) Two-sided t-test")
t_54 <- t.test(temp_int, mu = params_54$mu0, alternative = "two.sided",
               conf.level = 0.95)
xbar_54 <- mean(temp_int)
s_54    <- sd(temp_int)
se_54   <- s_54 / sqrt(params_54$n)
t0_54   <- (xbar_54 - params_54$mu0) / se_54
p_54    <- 2 * (1 - pt(abs(t0_54), df = params_54$n - 1))

cat("H0: mu  = 22.5\n")
cat("H1: mu != 22.5\n")
cat(sprintf("xbar     = %.4f\n", xbar_54))
cat(sprintf("s        = %.4f\n", s_54))
cat(sprintf("t0       = %.4f\n", t0_54))
cat(sprintf("df       = %d\n",   params_54$n - 1))
cat(sprintf("P-value  = %.4f\n", p_54))
cat("Decision: ",
    if (p_54 < params_54$alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")
cat(sprintf("[t.test cross-check]  t = %.4f, p = %.4f\n",
```

```r
            t_54$statistic, t_54$p.value))

# (c) 95% CI
print_subhead("(c) 95% CI for mean interior temperature")
CI_54 <- as.numeric(t_54$conf.int)
cat(sprintf("CI = (%.4f , %.4f)\n", CI_54[1], CI_54[2]))

# (d) 樣本數，要 power >= 0.9 偵測 mu=22.75
print_subhead("(d) Sample size for power >= 0.9, true mu = 22.75")
delta_54d  <- 22.75 - params_54$mu0
sigma_54d  <- s_54
n_req_54d  <- find_min_n_t(delta = delta_54d, sigma = sigma_54d,
                           alpha = params_54$alpha,
                           target_power = 0.90, alt = "two.sided")
pw_minus54 <- Pwr.t.test.custom(n_req_54d - 1, delta_54d, sigma_54d,
                                params_54$alpha, alt = "two.sided")
pw_at_54   <- Pwr.t.test.custom(n_req_54d,     delta_54d, sigma_54d,
                                params_54$alpha, alt = "two.sided")
cat(sprintf("delta       = %.4f, sigma (=s) = %.4f\n", delta_54d, sigma_54d))
cat(sprintf("Required n  = %d\n", n_req_54d))
cat(sprintf("Power at n-1 (%d) = %.4f\n", n_req_54d - 1, pw_minus54))
cat(sprintf("Power at n   (%d) = %.4f\n", n_req_54d,     pw_at_54))
#>
#> ====================================================
#>  EXERCISE 4-54: INTERIOR TEMPERATURE
#> ====================================================
#>
#> --- Summary statistics ---
#>   n   Mean    Var Std_Dev Std_Error Median
#> 1 5 22.496 0.1431  0.3783    0.1692  22.59
#>
#> --- (b) Normality check  (Shapiro-Wilk + Q-Q plot) ---
#>
#>  Shapiro-Wilk normality test
#>
#> data:  temp_int
#> W = 0.95969, p-value = 0.8058
#>
#> 樣本數很小，Shapiro-Wilk power 不高，需要搭配 Q-Q plot 一起看。
#> Plots saved: plots/ex4_54_qq.png, plots/ex4_54_hist.png
#>
#> --- (a) Two-sided t-test ---
#> H0: mu  = 22.5
#> H1: mu != 22.5
#> xbar     = 22.4960
#> s        = 0.3783
#> t0       = -0.0236
#> df       = 4
#> P-value  = 0.9823
#> Decision:  FAIL TO REJECT H0
#> [t.test cross-check]  t = -0.0236, p = 0.9823
#>
#> --- (c) 95% CI for mean interior temperature ---
#> CI = (22.0262 , 22.9658)
```

```
#>
#> --- (d) Sample size for power >= 0.9, true mu = 22.75 ---
#> delta       = 0.2500, sigma (=s) = 0.3783
#> Required n  = 27
#> Power at n-1 (26) = 0.8993
#> Power at n   (27) = 0.9106
```

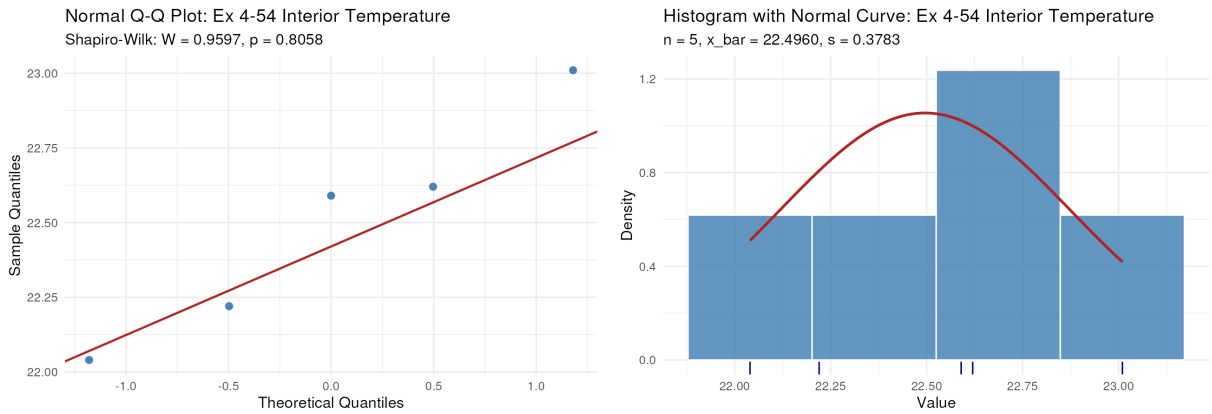

Figure 2: Ex 4-54 常態性檢查：Q-Q plot 與直方圖（已標註 Shapiro-Wilk 結果與描述統計）

最終答案 Final Answer

- **(a)** $t_0 = -0.0236$、P-value $= 0.9823$，遠大於 $\alpha = 0.05$，**FAIL TO REJECT** H_0：沒有夠強的證據說 $\mu \neq 22.5$。
- **(b)** Shapiro-Wilk $W = 0.9597$、$p = 0.8058$；Q-Q plot 上 5 個點落在斜線附近，常態假設可以接受。樣本數很小，不過度宣稱。
- **(c)** 95% CI for μ：$(22.0262,\ 22.9658)$，22.5 落在區間裡，跟 (a) 一致。
- **(d)** 用 sample SD 當 σ 估計，要 power ≥ 0.9 偵測 $\mu = 22.75$，$n = \mathbf{27}$（$n = 26$ 時 power $= 0.8993$ 還差一點，$n = 27$ 時 power $= 0.9106$ 過關）。

Exercise 4-59: Diode Breakdown Voltage

> **4-59.** In building electronic circuitry, the breakdown voltage of diodes is an important quality characteristic. The breakdown voltage of 12 diodes was recorded as follows:
> 9.099, 9.174, 9.327, 9.377, 8.471, 9.575, 9.514, 8.928, 8.800, 8.920, 9.913, and 8.306.
> (a) Check the normality assumption for the data.
> (b) Test the claim that the mean breakdown voltage is less than 9 volts with a significance level of 0.05.
> (c) Construct a 95% one-sided upper confidence bound on the mean breakdown voltage.
> (d) Use the bound found in part (c) to test the hypothesis.
> (e) Suppose that the true breakdown voltage is 8.8 volts; it is important to detect this with a probability of at least 0.95. Using the sample standard deviation to estimate the population standard deviation and a significance level of 0.05, determine the necessary sample size.

```r
# ==============================================================================
# EXERCISE 4-59: Diode Breakdown Voltage (Section 4-5)
# n=12, mu0=9, alpha=0.05, H1: mu < 9
# ==============================================================================
print_header("EXERCISE 4-59: DIODE BREAKDOWN VOLTAGE")

diode_v <- c(9.099, 9.174, 9.327, 9.377, 8.471, 9.575,
             9.514, 8.928, 8.800, 8.920, 9.913, 8.306)

params_59 <- list(mu0 = 9, alpha = 0.05, n = length(diode_v))

print_subhead("Summary statistics")
print(get_stats(diode_v))

# (a) 常態性
print_subhead("(a) Normality check")
(sw_59 <- shapiro.test(diode_v))
save_normality_plots(diode_v, "Ex 4-59 Diode Voltage", "ex4_59")
cat("Plots saved: plots/ex4_59_qq.png, plots/ex4_59_hist.png\n")

# (b) t 檢定，one-sided less
print_subhead("(b) One-sided t-test, H1: mu < 9")
t_59      <- t.test(diode_v, mu = params_59$mu0, alternative = "less",
                    conf.level = 0.95)
xbar_59   <- mean(diode_v)
s_59      <- sd(diode_v)
se_59     <- s_59 / sqrt(params_59$n)
t0_59     <- (xbar_59 - params_59$mu0) / se_59
p_59      <- pt(t0_59, df = params_59$n - 1)

cat(sprintf("xbar    = %.4f\n", xbar_59))
cat(sprintf("s       = %.4f\n", s_59))
cat(sprintf("t0      = %.4f\n", t0_59))
cat(sprintf("df      = %d\n",   params_59$n - 1))
cat(sprintf("P-value = %.4f\n", p_59))
cat("Decision: ",
    if (p_59 < params_59$alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")
cat(sprintf("[t.test cross-check]  t = %.4f, p = %.4f\n",
            t_59$statistic, t_59$p.value))
```

```r
# (c) 95% one-sided upper bound
print_subhead("(c) 95% one-sided upper confidence bound")
UCL_59 <- xbar_59 + qt(1 - params_59$alpha, df = params_59$n - 1) * se_59
cat(sprintf("UCL = %.4f\n", UCL_59))
cat(sprintf("CI : ( -Inf , %.4f ]\n", UCL_59))

# (d) 用 UCL 做檢定
print_subhead("(d) Test using the upper bound from (c)")
cat(sprintf("UCL = %.4f, mu0 = %d => ", UCL_59, params_59$mu0))
if (UCL_59 < params_59$mu0) {
  cat("UCL < 9, mu0 lies outside the CI, reject H0.\n")
} else {
  cat("UCL >= 9, mu0 lies inside the CI, do not reject H0.\n")
}

# (e) 樣本數要 power >= 0.95, 偵測 mu=8.8
print_subhead("(e) Sample size for power >= 0.95, true mu = 8.8")
delta_59e <- 8.8 - params_59$mu0
sigma_59e <- s_59
n_req_59e <- find_min_n_t(delta = delta_59e, sigma = sigma_59e,
                          alpha = params_59$alpha,
                          target_power = 0.95, alt = "less")
pw_minus59 <- Pwr.t.test.custom(n_req_59e - 1, delta_59e, sigma_59e,
                                params_59$alpha, alt = "less")
pw_at_59   <- Pwr.t.test.custom(n_req_59e,     delta_59e, sigma_59e,
                                params_59$alpha, alt = "less")
cat(sprintf("delta = %.4f, sigma (=s) = %.4f\n", delta_59e, sigma_59e))
cat(sprintf("Required n = %d\n", n_req_59e))
cat(sprintf("Power at n-1 (%d) = %.4f\n", n_req_59e - 1, pw_minus59))
cat(sprintf("Power at n   (%d) = %.4f\n", n_req_59e,     pw_at_59))
#>
#> ====================================================
#>  EXERCISE 4-59: DIODE BREAKDOWN VOLTAGE
#> ====================================================
#>
#> --- Summary statistics ---
#>    n  Mean    Var Std_Dev Std_Error Median
#> 1 12 9.117 0.2151  0.4638    0.1339 9.1365
#>
#> --- (a) Normality check ---
#>
#>  Shapiro-Wilk normality test
#>
#> data:  diode_v
#> W = 0.98423, p-value = 0.9953
#>
#> Plots saved: plots/ex4_59_qq.png, plots/ex4_59_hist.png
#>
#> --- (b) One-sided t-test, H1: mu < 9 ---
#> xbar    = 9.1170
#> s       = 0.4638
#> t0      = 0.8738
#> df      = 11
#> P-value = 0.7996
```

```
#> Decision:  FAIL TO REJECT H0
#> [t.test cross-check]  t = 0.8738, p = 0.7996
#>
#> --- (c) 95% one-sided upper confidence bound ---
#> UCL = 9.3575
#> CI : ( -Inf , 9.3575 ]
#>
#> --- (d) Test using the upper bound from (c) ---
#> UCL = 9.3575, mu0 = 9 => UCL >= 9, mu0 lies inside the CI, do not reject H0.
#>
#> --- (e) Sample size for power >= 0.95, true mu = 8.8 ---
#> delta = -0.2000, sigma (=s) = 0.4638
#> Required n = 60
#> Power at n-1 (59) = 0.9483
#> Power at n   (60) = 0.9512
```

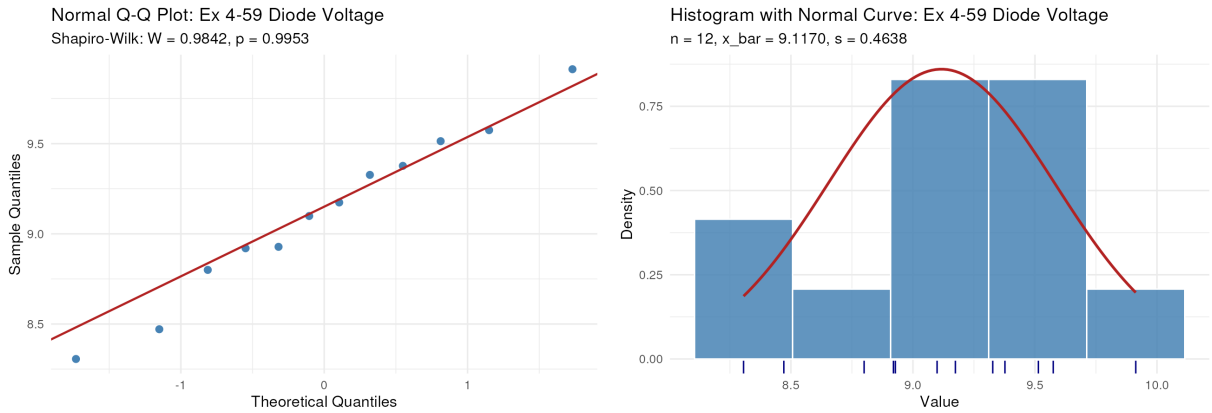

Figure 3: Ex 4-59 常態性檢查：Q-Q plot 與直方圖

最終答案 Final Answer

- **(a)** Shapiro-Wilk $W = 0.9842$、$p = 0.9953$；Q-Q plot 與直方圖看起來都接近常態，沒有明顯偏離。
- **(b)** $t_0 = 0.8738$、P-value $= 0.7996$，**FAIL TO REJECT** H_0：沒有證據支持「平均崩潰電壓低於 9 V」這個說法。
- **(c)** 95% 一邊上界 CI：$(-\infty,\ 9.3575]$。
- **(d)** UCL $= 9.3575$ 比 9 大，9 還在 CI 裡 $\Rightarrow$ 不拒絕 H_0（跟 (b) 一致）。
- **(e)** 用 sample SD 估 σ，要在 $\alpha = 0.05$ 下偵測 $\mu = 8.8$ 的機率達 0.95，$n = \mathbf{60}$（$n = 59$ 時 power $= 0.9483$ 差一點，$n = 60$ 時 power $= 0.9512$ 剛好過）。

Exercise 4-65: Cloud Seeding Rainfall

> **4-65.** Cloud seeding has been studied for many decades as a weather modification procedure (for an interesting study of this subject, see the article in *Technometrics*, "A Bayesian Analysis of a Multiplicative Treatment Effect in Weather Modification," Vol. 17, 1975, pp. 161-166). The rainfall in acre-feet from 20 clouds that were selected at random and seeded with silver nitrate follows:
> 18.0, 30.7, 19.8, 27.1, 22.3, 18.8, 31.8, 23.4, 21.2, 27.9, 31.9, 27.1, 25.0, 24.7, 26.9, 21.8, 29.2, 34.8, 26.7, and 31.6.
>
> (a) Can you support a claim that mean rainfall from seeded clouds exceeds 25 acre-feet? Use $\alpha = 0.01$. Find the P-value.
> (b) Check that rainfall is normally distributed.
> (c) Compute the power of the test if the true mean rainfall is 27 acre-feet.
> (d) What sample size would be required to detect a true mean rainfall of 27.5 acre-feet if we wanted the power of the test to be at least 0.9?
> (e) Explain how the question in part (a) could be answered by constructing a one-sided confidence bound on the mean diameter.

```r
# ==============================================================================
# EXERCISE 4-65: Cloud Seeding Rainfall (Section 4-5)
# n=20, mu0=25, alpha=0.01, H1: mu > 25
# Note: 題目 (e) 寫 mean diameter，但全題討論的是 rainfall，疑似誤植
# ==============================================================================
print_header("EXERCISE 4-65: CLOUD SEEDING RAINFALL")

rain <- c(18.0, 30.7, 19.8, 27.1, 22.3, 18.8, 31.8, 23.4, 21.2, 27.9,
          31.9, 27.1, 25.0, 24.7, 26.9, 21.8, 29.2, 34.8, 26.7, 31.6)

params_65 <- list(mu0 = 25, alpha = 0.01, n = length(rain))

print_subhead("Summary statistics")
print(get_stats(rain))

# (b) 常態性 (先看圖再做 t test)
print_subhead("(b) Normality check")
(sw_65 <- shapiro.test(rain))
save_normality_plots(rain, "Ex 4-65 Cloud Seeding Rainfall", "ex4_65")
cat("Plots saved: plots/ex4_65_qq.png, plots/ex4_65_hist.png\n")

# (a) t 檢定，one-sided greater
print_subhead("(a) One-sided t-test, P-value")
t_65    <- t.test(rain, mu = params_65$mu0, alternative = "greater",
                  conf.level = 0.99)
xbar_65 <- mean(rain)
s_65    <- sd(rain)
se_65   <- s_65 / sqrt(params_65$n)
t0_65   <- (xbar_65 - params_65$mu0) / se_65
p_65    <- 1 - pt(t0_65, df = params_65$n - 1)
cat(sprintf("xbar     = %.4f\n", xbar_65))
cat(sprintf("s        = %.4f\n", s_65))
cat(sprintf("t0       = %.4f\n", t0_65))
cat(sprintf("df       = %d\n",   params_65$n - 1))
cat(sprintf("P-value  = %.4f\n", p_65))
cat("Decision: ",
    if (p_65 < params_65$alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")
```

```r
cat(sprintf("[t.test cross-check]  t = %.4f, p = %.4f\n",
            t_65$statistic, t_65$p.value))

# (c) Power, true mu = 27
print_subhead("(c) Power if true mu = 27")
delta_65c <- 27 - params_65$mu0
pw_65c    <- Pwr.t.test.custom(params_65$n, delta_65c, s_65,
                               params_65$alpha, alt = "greater")
cat(sprintf("delta = %.4f, sigma (=s) = %.4f\n", delta_65c, s_65))
cat(sprintf("Power = %.4f, beta = %.4f\n", pw_65c, 1 - pw_65c))

# (d) 樣本數，要 power >= 0.9，true mu = 27.5
print_subhead("(d) Sample size for power >= 0.9, true mu = 27.5")
delta_65d  <- 27.5 - params_65$mu0
n_req_65d  <- find_min_n_t(delta = delta_65d, sigma = s_65,
                           alpha = params_65$alpha,
                           target_power = 0.90, alt = "greater")
pw_minus65 <- Pwr.t.test.custom(n_req_65d - 1, delta_65d, s_65,
                               params_65$alpha, alt = "greater")
pw_at_65   <- Pwr.t.test.custom(n_req_65d,     delta_65d, s_65,
                               params_65$alpha, alt = "greater")
cat(sprintf("Required n = %d\n", n_req_65d))
cat(sprintf("Power at n-1 (%d) = %.4f\n", n_req_65d - 1, pw_minus65))
cat(sprintf("Power at n   (%d) = %.4f\n", n_req_65d,     pw_at_65))

# (e) 一邊下界 CI 解釋
# 題目寫 mean diameter，但脈絡是 rainfall，這邊照題目敘述但加註
print_subhead("(e) One-sided lower confidence bound for mean rainfall")
cat("Note: The textbook says 'mean diameter' in part (e); the context indicates ")
cat("this should be 'mean rainfall'. We answer for mean rainfall.\n")
LCL_65e <- xbar_65 - qt(1 - params_65$alpha, df = params_65$n - 1) * se_65
cat(sprintf("LCL = %.4f\n", LCL_65e))
cat(sprintf("CI : [ %.4f , Inf )\n", LCL_65e))
cat(sprintf("Compare LCL with mu0 = %d: ", params_65$mu0))
if (LCL_65e > params_65$mu0) {
  cat("LCL > 25, reject H0 -- supports the claim.\n")
} else {
  cat("LCL <= 25, do not reject H0.\n")
}
#>
#> ==================================================
#>  EXERCISE 4-65: CLOUD SEEDING RAINFALL
#> ==================================================
#>
#> --- Summary statistics ---
#>    n   Mean    Var Std_Dev Std_Error Median
#> 1 20 26.035 22.894  4.7848    1.0699   26.8
#>
#> --- (b) Normality check ---
#>
#>  Shapiro-Wilk normality test
#>
#> data:  rain
#> W = 0.97025, p-value = 0.7601
```

```
#>
#> Plots saved: plots/ex4_65_qq.png, plots/ex4_65_hist.png
#>
#> --- (a) One-sided t-test, P-value ---
#> xbar      = 26.0350
#> s         = 4.7848
#> t0        = 0.9674
#> df        = 19
#> P-value   = 0.1728
#> Decision:  FAIL TO REJECT H0
#> [t.test cross-check]  t = 0.9674, p = 0.1728
#>
#> --- (c) Power if true mu = 27 ---
#> delta = 2.0000, sigma (=s) = 4.7848
#> Power = 0.2781, beta = 0.7219
#>
#> --- (d) Sample size for power >= 0.9, true mu = 27.5 ---
#> Required n = 51
#> Power at n-1 (50) = 0.8971
#> Power at n   (51) = 0.9037
#>
#> --- (e) One-sided lower confidence bound for mean rainfall ---
#> Note: The textbook says 'mean diameter' in part (e); the context indicates this should
↪  be 'mean rainfall'. We answer for mean rainfall.
#> LCL = 23.3180
#> CI : [ 23.3180 , Inf )
#> Compare LCL with mu0 = 25: LCL <= 25, do not reject H0.
```

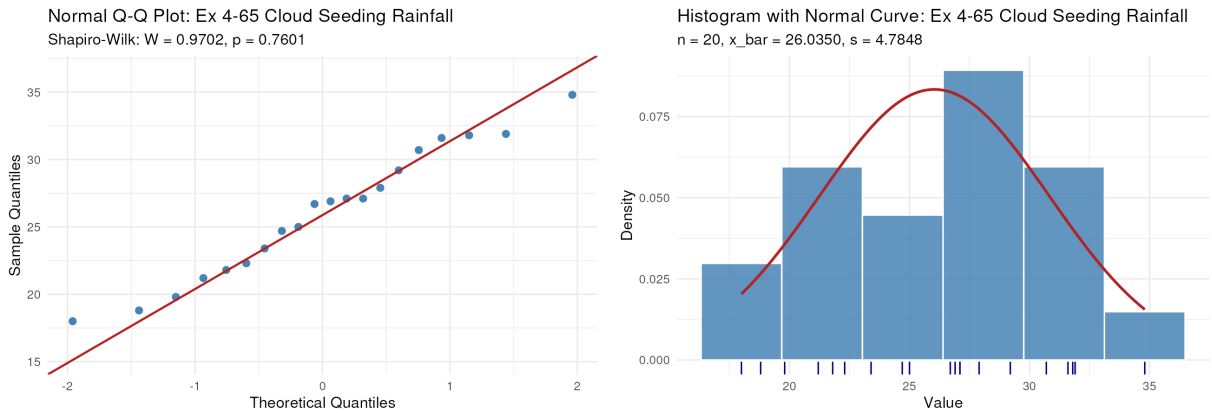

Figure 4: Ex 4-65 常態性檢查：Q-Q plot 與直方圖

最終答案 Final Answer

- **(a)** $t_0 = 0.9674$、P-value $= 0.1728$，$\alpha = 0.01$ 下 **FAIL TO REJECT** H_0：證據還不夠強，沒辦法支持「平均雨量超過 25 acre-feet」。
- **(b)** Shapiro-Wilk $W = 0.9702$、$p = 0.7601$，常態假設可以接受。
- **(c)** 真實 $\mu = 27$ 時 power ≈ 0.2781、$\beta \approx 0.7219$，這代表 $n = 20$ 對 2 acre-feet 的差距偵測力其實不高。
- **(d)** 想偵測 $\mu = 27.5$、power ≥ 0.9，$n =$ **51** ($n = 50$ 時 power $= 0.8971$，$n = 51$ 過關)。
- **(e)** 95% 一邊下界 CI (脈絡上看是 mean rainfall)：$[23.3180, \infty)$。LCL 沒跑到 25 上面，跟 (a) 一樣 不拒絕 H_0；用一邊下界比較 25 跟 P-value 的結論完全一致。
- 補註：題目英文寫 *"mean diameter"* 跟前後文的雨量主題不符，疑似原書誤植。本題答案依雨量脈絡作答。

Exercise 4-71: Inference on σ (Titanium Alloy)

> **4-71.** The percentage of titanium in an alloy used in aerospace castings is measured in 51 randomly selected parts. The sample standard deviation is $s = 0.37$.
>
> (a) Test the hypothesis $H_0 : \sigma = 0.35$ versus $H_1 : \sigma \neq 0.35$ using $\alpha = 0.05$. State any necessary assumptions about the underlying distribution of the data.
> (b) Find the P-value for this test.
> (c) Construct a 95% two-sided CI for σ.
> (d) Use the CI in part (c) to test the hypothesis.

```r
# ==============================================================================
# EXERCISE 4-71: Titanium Alloy Standard Deviation (Section 4-6)
# n=51, s=0.37, sigma0=0.35, alpha=0.05, two-sided
# ==============================================================================
print_header("EXERCISE 4-71: TITANIUM ALLOY SIGMA TEST")

# 題目只給 summary，沒有原始資料，hard-code 即可
n_71      <- 51
s_71      <- 0.37
sigma0_71 <- 0.35
alpha_71  <- 0.05
df_71     <- n_71 - 1

cat(" 假設母體近似常態分配，這個 chi-square test 才有效。\n")

# (a)(b) 卡方統計量、P-value
print_subhead("(a)(b) Chi-square test")
chi0_71   <- (n_71 - 1) * s_71^2 / sigma0_71^2
# 雙尾 P-value 取 2 * min(left, right)
p_left    <- pchisq(chi0_71, df_71)
p_right   <- 1 - pchisq(chi0_71, df_71)
p_71      <- 2 * min(p_left, p_right)

cat("H0: sigma  = 0.35\n")
cat("H1: sigma != 0.35\n")
cat(sprintf("chi0    = %.4f\n", chi0_71))
cat(sprintf("df      = %d\n",   df_71))
cat(sprintf("P-value = %.4f\n", p_71))
cat(sprintf("Critical: chi^2_{0.025,%d} = %.4f, chi^2_{0.975,%d} = %.4f\n",
            df_71, qchisq(0.975, df_71), df_71, qchisq(0.025, df_71)))
cat("Decision: ",
    if (p_71 < alpha_71) "REJECT H0" else "FAIL TO REJECT H0", "\n")

# (c) 95% CI for sigma
# 先做 sigma^2 的 CI 再開根號
print_subhead("(c) 95% two-sided CI for sigma")
LCL_var <- (n_71 - 1) * s_71^2 / qchisq(1 - alpha_71/2, df_71)
UCL_var <- (n_71 - 1) * s_71^2 / qchisq(    alpha_71/2, df_71)
LCL_sd  <- sqrt(LCL_var)
UCL_sd  <- sqrt(UCL_var)
cat(sprintf("CI for sigma^2 : (%.4f , %.4f)\n", LCL_var, UCL_var))
cat(sprintf("CI for sigma   : (%.4f , %.4f)\n", LCL_sd,  UCL_sd))

# (d) 用 CI 做檢定
```

```r
print_subhead("(d) Test the hypothesis using the CI")
cat(sprintf("Is sigma0 = %.2f inside (%.4f, %.4f)? ", sigma0_71, LCL_sd, UCL_sd))
if (sigma0_71 >= LCL_sd && sigma0_71 <= UCL_sd) {
  cat("Yes => do not reject H0.\n")
} else {
  cat("No  => reject H0.\n")
}
#>
#> ====================================================
#>  EXERCISE 4-71: TITANIUM ALLOY SIGMA TEST
#> ====================================================
#> 假設母體近似常態分配，這個 chi-square test 才有效。
#>
#> --- (a)(b) Chi-square test ---
#> H0: sigma  = 0.35
#> H1: sigma != 0.35
#> chi0     = 55.8776
#> df       = 50
#> P-value = 0.5272
#> Critical: chi^2_{0.025,50} = 71.4202, chi^2_{0.975,50} = 32.3574
#> Decision:  FAIL TO REJECT H0
#>
#> --- (c) 95% two-sided CI for sigma ---
#> CI for sigma^2 : (0.0958 , 0.2115)
#> CI for sigma   : (0.3096 , 0.4599)
#>
#> --- (d) Test the hypothesis using the CI ---
#> Is sigma0 = 0.35 inside (0.3096, 0.4599)? Yes => do not reject H0.
```

最終答案 Final Answer

- 前提：母體要近似常態，這個 χ^2 推論才有效。
- **(a)(b)** $\chi_0^2 = (n-1)s^2/\sigma_0^2 = 55.8776$、$\mathrm{df} = 50$，雙尾 P-value $= 0.5272 \Rightarrow$ **FAIL TO REJECT** H_0。
- **(c)** 先算 σ^2 的 95% CI：$(0.0958,\ 0.2115)$，開根號得 σ 的 95% CI：$(0.3096,\ 0.4599)$。
- **(d)** $\sigma_0 = 0.35$ 落在 CI $(0.3096, 0.4599)$ 內 $\Rightarrow$ 不拒絕 H_0，跟 (a) 一致。

Exercise 4-75: Van Rollover Proportion

4-75. Large passenger vans are thought to have a high propensity of rollover accidents when fully loaded. Thirty accidents of these vans were examined, and 11 vans had rolled over.

(a) Test the claim that the proportion of rollovers exceeds 0.25 with $\alpha = 0.10$.
(b) Suppose that the true $p = 0.35$ and $\alpha = 0.10$. What is the β-error for this test?
(c) Suppose that the true $p = 0.35$ and $\alpha = 0.10$. How large a sample would be required if we want $\beta = 0.10$?
(d) Find a 90% traditional lower confidence bound on the rollover rate of these vans.
(e) Use the confidence bound found in part (d) to test the hypothesis.
(f) How large a sample would be required to be at least 95% confident that the error on p is less than 0.02? Use an initial estimate of p from this problem.

```r
# ==============================================================================
# EXERCISE 4-75: Van Rollover Proportion (Section 4-7)
# n=30, x=11, p0=0.25, alpha=0.10, H1: p > 0.25
# ==============================================================================
print_header("EXERCISE 4-75: VAN ROLLOVER PROPORTION")

n_75    <- 30
x_75    <- 11
p0_75   <- 0.25
alpha_75 <- 0.10
phat_75 <- x_75 / n_75

cat(sprintf("phat = %d / %d = %.4f\n", x_75, n_75, phat_75))

# (a) z 檢定
print_subhead("(a) One-proportion z-test, H1: p > 0.25")
se0_75 <- sqrt(p0_75 * (1 - p0_75) / n_75)
z0_75  <- (phat_75 - p0_75) / se0_75
p_75   <- 1 - pnorm(z0_75)
cat(sprintf("z0      = %.4f\n", z0_75))
cat(sprintf("P-value = %.4f\n", p_75))
cat("Decision: ",
    if (p_75 < alpha_75) "REJECT H0" else "FAIL TO REJECT H0", "\n")
# 套件交叉驗證 (correct=FALSE 不做連續性修正)
ptest_75 <- prop.test(x_75, n_75, p = p0_75, alternative = "greater",
                      correct = FALSE)
cat(sprintf("[prop.test cross-check]  chi^2 = %.4f, p = %.4f\n",
            ptest_75$statistic, ptest_75$p.value))

# (b) Beta，true p = 0.35
print_subhead("(b) Beta when true p = 0.35")
# 在 H0 下找臨界 phat
phat_crit <- p0_75 + qnorm(1 - alpha_75) * se0_75
true_p    <- 0.35
se_true   <- sqrt(true_p * (1 - true_p) / n_75)
beta_75b  <- pnorm((phat_crit - true_p) / se_true)
cat(sprintf("phat critical = %.4f\n", phat_crit))
cat(sprintf("beta          = %.4f\n", beta_75b))

# (c) 樣本數要 beta = 0.10
print_subhead("(c) Sample size so beta = 0.10 when true p = 0.35")
```

```r
z_alpha <- qnorm(1 - alpha_75)
z_beta  <- qnorm(1 - 0.10)
num_75  <- z_alpha * sqrt(p0_75*(1-p0_75)) + z_beta * sqrt(true_p*(1-true_p))
n_75c   <- ceiling((num_75 / (true_p - p0_75))^2)
cat(sprintf("Required n = %d\n", n_75c))

# (d) 90% one-sided lower bound
print_subhead("(d) 90% traditional one-sided lower confidence bound")
LCB_75 <- prop_wald_ci(x_75, n_75, conf_level = 0.90, side = "lower")[1]
cat(sprintf("LCB = %.4f\n", LCB_75))
cat(sprintf("CI : [ %.4f , 1 ]\n", LCB_75))

# (e) 用 LCB 做檢定
print_subhead("(e) Test using the lower bound")
cat(sprintf("LCB = %.4f vs p0 = %.2f => ", LCB_75, p0_75))
if (LCB_75 > p0_75) {
  cat("LCB > 0.25, reject H0 -- claim supported.\n")
} else {
  cat("LCB <= 0.25, do not reject H0.\n")
}

# (f) 樣本數要誤差 < 0.02，95% confident，用 phat = 11/30
print_subhead("(f) Sample size for E < 0.02, 95% confidence, phat = 11/30")
z_95   <- qnorm(0.975)
n_75f  <- ceiling(z_95^2 * phat_75 * (1 - phat_75) / 0.02^2)
cat(sprintf("Using phat = %.4f, required n = %d\n", phat_75, n_75f))
#>
#> ==================================================
#>  EXERCISE 4-75: VAN ROLLOVER PROPORTION
#> ==================================================
#> phat = 11 / 30 = 0.3667
#>
#> --- (a) One-proportion z-test, H1: p > 0.25 ---
#> z0      = 1.4757
#> P-value = 0.0700
#> Decision:  REJECT H0
#> [prop.test cross-check]  chi^2 = 2.1778, p = 0.0700
#>
#> --- (b) Beta when true p = 0.35 ---
#> phat critical = 0.3513
#> beta          = 0.5060
#>
#> --- (c) Sample size so beta = 0.10 when true p = 0.35 ---
#> Required n = 136
#>
#> --- (d) 90% traditional one-sided lower confidence bound ---
#> LCB = 0.2539
#> CI : [ 0.2539 , 1 ]
#>
#> --- (e) Test using the lower bound ---
#> LCB = 0.2539 vs p0 = 0.25 => LCB > 0.25, reject H0 -- claim supported.
#>
#> --- (f) Sample size for E < 0.02, 95% confidence, phat = 11/30 ---
#> Using phat = 0.3667, required n = 2231
```

最終答案 Final Answer

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- **(a)** $z_0 = 1.4757$、P-value $= 0.0700$，$\alpha = 0.10$ 下 **REJECT** H_0；「翻覆比例超過 0.25」這個論點站得住腳。
- **(b)** 真實 $p = 0.35$ 時 $\beta \approx 0.5060$（蠻高的，因為效應量只有 0.10、$n = 30$ 偏小）。

- **(f)** 用 $\hat{p} = 11/30$、要 95% 信心、誤差 < 0.02，所需 $n = \mathbf{2231}$。

Exercise 4-76: Helmet Damage Proportion

4-76. A random sample of 50 suspension helmets used by motorcycle riders and automobile race-car drivers was subjected to an impact test, and on 18 of these helmets some damage was observed.

(a) Test the hypotheses $H_0 : p = 0.3$ versus $H_1 : p \neq 0.3$ with $\alpha = 0.05$.

(b) Find the P-value for this test.

(c) Find a 95% two-sided traditional CI on the true proportion of helmets of this type that would show damage from this test. Explain how this confidence interval can be used to test the hypothesis in part (a).

(d) Using the point estimate of p obtained from the preliminary sample of 50 helmets, how many helmets must be tested to be 95% confident that the error in estimating the true value of p is less than 0.02?

(e) How large must the sample be if we wish to be at least 95% confident that the error in estimating p is less than 0.02, regardless of the true value of p?

```r
# ==============================================================================
# EXERCISE 4-76: Helmet Damage Proportion (Section 4-7)
# n=50, x=18, p0=0.3, alpha=0.05, H1: p != 0.3
# ==============================================================================
print_header("EXERCISE 4-76: HELMET DAMAGE PROPORTION")

n_76    <- 50
x_76    <- 18
p0_76   <- 0.30
alpha_76 <- 0.05
phat_76 <- x_76 / n_76

cat(sprintf("phat = %d / %d = %.4f\n", x_76, n_76, phat_76))

# (a)(b) z 檢定 + P-value
print_subhead("(a)(b) Two-sided z-test")
se0_76 <- sqrt(p0_76 * (1 - p0_76) / n_76)
z0_76  <- (phat_76 - p0_76) / se0_76
p_76   <- 2 * (1 - pnorm(abs(z0_76)))
cat(sprintf("z0      = %.4f\n", z0_76))
cat(sprintf("P-value = %.4f\n", p_76))
cat("Decision: ",
    if (p_76 < alpha_76) "REJECT H0" else "FAIL TO REJECT H0", "\n")
# 套件交叉驗證
ptest_76 <- prop.test(x_76, n_76, p = p0_76, alternative = "two.sided",
                      correct = FALSE)
cat(sprintf("[prop.test cross-check]  chi^2 = %.4f, p = %.4f\n",
            ptest_76$statistic, ptest_76$p.value))

# (c) 95% two-sided traditional CI
print_subhead("(c) 95% two-sided traditional CI")
CI_76 <- prop_wald_ci(x_76, n_76, conf_level = 0.95, side = "two.sided")
cat(sprintf("CI = (%.4f , %.4f)\n", CI_76[1], CI_76[2]))
cat(sprintf("Is p0 = %.2f inside the CI? ", p0_76))
if (p0_76 >= CI_76[1] && p0_76 <= CI_76[2]) {
  cat("Yes => fail to reject H0 (consistent with the test).\n")
} else {
  cat("No  => reject H0.\n")
}
```

```r
# (d) 樣本數要 E < 0.02，用 phat = 18/50
print_subhead("(d) Sample size for E < 0.02, 95% conf., using phat = 18/50")
z95_76 <- qnorm(0.975)
n_76d  <- ceiling(z95_76^2 * phat_76 * (1 - phat_76) / 0.02^2)
cat(sprintf("Using phat = %.4f, required n = %d\n", phat_76, n_76d))

# (e) 保守樣本數，用 p = 0.5
print_subhead("(e) Conservative sample size using p = 0.5")
n_76e <- ceiling(z95_76^2 * 0.5 * 0.5 / 0.02^2)
cat(sprintf("Using p = 0.50, required n = %d\n", n_76e))
#>
#> ==================================================
#>  EXERCISE 4-76: HELMET DAMAGE PROPORTION
#> ==================================================
#> phat = 18 / 50 = 0.3600
#>
#> --- (a)(b) Two-sided z-test ---
#> z0      = 0.9258
#> P-value = 0.3545
#> Decision:  FAIL TO REJECT H0
#> [prop.test cross-check]  chi^2 = 0.8571, p = 0.3545
#>
#> --- (c) 95% two-sided traditional CI ---
#> CI = (0.2270 , 0.4930)
#> Is p0 = 0.30 inside the CI? Yes => fail to reject H0 (consistent with the test).
#>
#> --- (d) Sample size for E < 0.02, 95% conf., using phat = 18/50 ---
#> Using phat = 0.3600, required n = 2213
#>
#> --- (e) Conservative sample size using p = 0.5 ---
#> Using p = 0.50, required n = 2401
```

最終答案 Final Answer

- **(a)(b)** $z_0 = 0.9258$、P-value $= 0.3545$，**FAIL TO REJECT** H_0：沒有夠強的證據說 $p \neq 0.3$。
- **(c)** 95% 雙尾傳統 (Wald) CI：$(0.2270,\ 0.4930)$；$p_0 = 0.30$ 落在區間裡 $\Rightarrow$ 不拒絕 H_0，跟 (a) 結論一致。這是 CI 跟雙尾檢定的等價性。
- **(d)** 用 $\hat{p} = 18/50$ 估算誤差 < 0.02，$n =$ **2213**。
- **(e)** 不知道 p 時用最保守的 $p = 0.5$，$n =$ **2401**。

Exercise 4-89: Agresti-Coull CI

> **4-89.** Consider the helmet data given in Exercise 4-76. Calculate the 95% Agresti-Coull two-sided CI from equation 4-76 and compare it to the traditional CI in the original exercise.

```r
# ==============================================================================
# EXERCISE 4-89: Agresti-Coull CI vs Traditional (Section 4-7)
# 用 4-76 的資料: n=50, x=18, 95% confidence
# ==============================================================================
print_header("EXERCISE 4-89: AGRESTI-COULL CI")

ac_89 <- agresti_coull_ci(x_76, n_76, conf_level = 0.95)

# Agresti-Coull 是把 (x, n) 換成 (x + z^2/2, n + z^2)
z_89   <- qnorm(0.975)
n_t_89 <- n_76 + z_89^2
p_t_89 <- (x_76 + z_89^2 / 2) / n_t_89

cat("Agresti-Coull 用調整過的樣本數和點估計來建立 CI，這是和傳統 Wald CI 的核心差別。\n")
cat(sprintf("Adjusted n_tilde = %.4f\n", n_t_89))
cat(sprintf("Adjusted p_tilde = %.4f\n", p_t_89))
cat(sprintf("Agresti-Coull CI : (%.4f , %.4f)\n", ac_89[1], ac_89[2]))
cat(sprintf("Traditional CI   : (%.4f , %.4f)\n", CI_76[1], CI_76[2]))
cat(sprintf("Width comparison : AC width = %.4f, traditional width = %.4f\n",
            ac_89[2] - ac_89[1], CI_76[2] - CI_76[1]))
#>
#> ====================================================
#>  EXERCISE 4-89: AGRESTI-COULL CI
#> ====================================================
#> Agresti-Coull 用調整過的樣本數和點估計來建立 CI，這是和傳統 Wald CI 的核心差別。
#> Adjusted n_tilde = 53.8415
#> Adjusted p_tilde = 0.3700
#> Agresti-Coull CI : (0.2410 , 0.4989)
#> Traditional CI   : (0.2270 , 0.4930)
#> Width comparison : AC width = 0.2579, traditional width = 0.2661
```

最終答案 Final Answer

- Agresti-Coull 的核心改動是把 (x, n) 換成 $(x + z^2/2,\ n + z^2)$ 後再套 Wald 形式。這裡 $\tilde{n} = 53.8415$、$\tilde{p} = 0.3700$（明顯偏向 0.5 一點點）。
- Agresti-Coull 95% CI：$(0.2410,\ 0.4989)$，寬度 0.2579。
- 4-76 的傳統 Wald 95% CI：$(0.2270,\ 0.4930)$，寬度 0.2661。
- 兩條 CI 看起來差距很小，但 Agresti-Coull 區間寬度略小、且中心略偏向 0.5；在 n 不大或 $\hat{p}$ 接近 0/1 時，它的實際涵蓋率比 Wald 更接近名目水準（這也是 Brown, Cai & DasGupta 推薦它的理由）。

Exercise 4-95: Prediction & Tolerance Interval

> **4-95.** Consider the breakdown voltage of diodes described in Exercise 4-59.
> (a) Construct a 99% PI for the breakdown voltage of a single diode.
> (b) Find a tolerance interval for the breakdown voltage that includes 99% of the diodes with 99% confidence.

```r
# ==============================================================================
# EXERCISE 4-95: Prediction & Tolerance Interval (Section 4-8)
# 沿用 4-59 的二極體資料，99% PI 與 99% / 99% TI
# ==============================================================================
print_header("EXERCISE 4-95: PI AND TOLERANCE INTERVAL")

# (a) 99% PI for one future observation
print_subhead("(a) 99% Prediction Interval for one future diode")
PI_95 <- prediction_interval_normal(diode_v, conf_level = 0.99)
cat(sprintf("xbar = %.4f, s = %.4f, n = %d\n",
            mean(diode_v), sd(diode_v), length(diode_v)))
cat(sprintf("PI = (%.4f , %.4f)\n", PI_95[1], PI_95[2]))
cat("PI 的標的是『下一筆觀測值』，比 CI 寬一些是正常的。\n")

# (b) Tolerance interval, 99% population coverage with 99% confidence
print_subhead("(b) 99%/99% two-sided normal tolerance interval")
TI_95 <- tolerance::normtol.int(diode_v, alpha = 0.01, P = 0.99, side = 2)
print(TI_95)
cat(sprintf("\nTI = (%.4f , %.4f)\n",
            TI_95$`2-sided.lower`, TI_95$`2-sided.upper`))
cat("TI 是希望涵蓋『母體中 99% 的個體』，而不是平均數，所以會比 PI 再寬一點。\n")

# 視覺化 PI / TI / CI 三者
ci_95 <- t.test(diode_v, conf.level = 0.99)$conf.int
intervals_df <- tibble(
  type  = c("99% CI for mean", "99% PI", "99% / 99% TI"),
  lower  = c(ci_95[1], PI_95[1], TI_95$`2-sided.lower`),
  upper  = c(ci_95[2], PI_95[2], TI_95$`2-sided.upper`)
) %>% mutate(width = upper - lower)
print(intervals_df)

p_intvl <- ggplot(intervals_df, aes(y = fct_inorder(type))) +
  geom_segment(aes(x = lower, xend = upper, yend = type),
               linewidth = 1.5, color = "steelblue") +
  geom_point(aes(x = lower), size = 3, color = "firebrick") +
  geom_point(aes(x = upper), size = 3, color = "firebrick") +
  geom_vline(xintercept = mean(diode_v), linetype = "dashed",
             color = "darkgreen") +
  labs(title = "Ex 4-95: CI, PI, and Tolerance Interval (diode voltage)",
       x = "Breakdown voltage", y = NULL) +
  theme_minimal(base_size = 12)
ggsave("plots/ex4_95_intervals.png", p_intvl,
       width = 7, height = 4, dpi = 200)
cat("Plot saved: plots/ex4_95_intervals.png\n")


# ==============================================================================
```

```
# 結束
# ==============================================================================
cat("\n=====================================================\n")
cat(" All exercises completed.\n")
cat("=====================================================\n")
#>
#> ====================================================
#>  EXERCISE 4-95: PI AND TOLERANCE INTERVAL
#> ====================================================
#>
#> --- (a) 99% Prediction Interval for one future diode ---
#> xbar = 9.1170, s = 0.4638, n = 12
#> PI = (7.6177 , 10.6163)
#> PI 的標的是『下一筆觀測值』，比 CI 寬一些是正常的。
#>
#> --- (b) 99%/99% two-sided normal tolerance interval ---
#>   alpha    P x.bar 2-sided.lower 2-sided.upper
#> 1  0.01 0.99 9.117      6.736194      11.49781
#>
#> TI = (6.7362 , 11.4978)
#> TI 是希望涵蓋『母體中 99% 的個體』，而不是平均數，所以會比 PI 再寬一點。
#> # A tibble: 3 x 4
#>   type            lower upper width
#>   <chr>           <dbl> <dbl> <dbl>
#> 1 99% CI for mean  8.70  9.53 0.832
#> 2 99% PI           7.62 10.6  3.00
#> 3 99% / 99% TI     6.74 11.5  4.76
#> Plot saved: plots/ex4_95_intervals.png
#>
#> ====================================================
#>  All exercises completed.
#> ====================================================
```

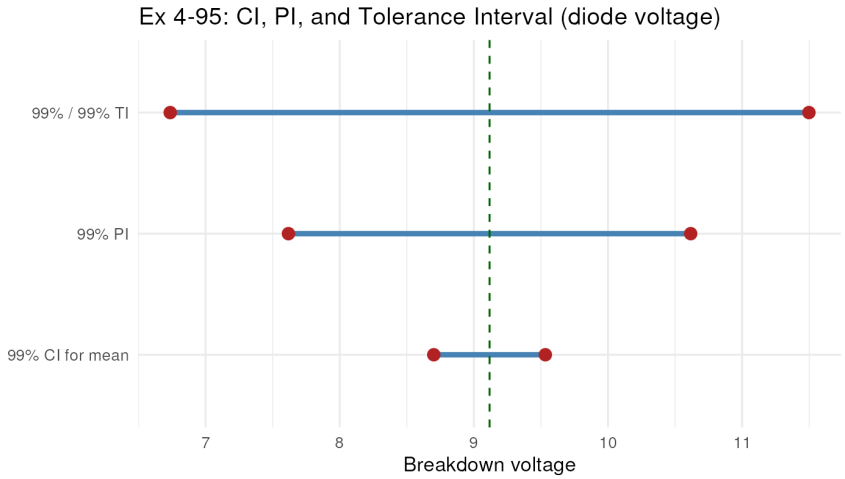

Figure 5: Ex 4-95 三種區間並列：99% CI for μ、99% PI、99%/99% TI

最終答案 Final Answer

- **(a)** 99% 預測區間 (PI) for 一筆未來觀測：(7.6177, 10.6163)。
- **(b)** 99% / 99% 常態容忍區間 (TI)：(6.7362, 11.4978)。
- 三種區間寬度順序剛好印證理論：CI for μ < PI < TI，因為覆蓋對象一個比一個廣（平均數 → 下一筆觀測 → 母體 99%）。

結語 / Closing Notes

做完這份作業，有幾個容易踩到雷、但這次都有刻意守住的地方順便整理一下：

- 資料原樣保留：題目給的數字一個都沒改，全部照原文 hard-code。
- 單尾 / 雙尾要看清楚：4-42、4-59、4-65、4-75 是單尾，4-54、4-71、4-76、4-89 是雙尾，方向搞錯就直接答錯。
- σ 已知 **vs** 未知：4-42、4-44 用 z；4-54、4-59、4-65 用 t（樣本數估算時用 sample SD 當 σ 的估計，題目有特別交代）。
- 常態假設：4-71 的 χ^2 推論明確寫了「假設母體近似常態」；4-54、4-59、4-65 的 t 檢定都附 Shapiro-Wilk 與 Q-Q plot 一起看，避免單看 Shapiro-Wilk 的低 power 結果做過度判斷。
- 樣本數：一律用 `ceiling()`，並且額外印 $n-1$ 與 n 兩點的 power 來驗證。
- **PI** 跟 **TI** 不能混：PI 是「下一筆觀測」的範圍，TI 是「母體覆蓋率」的範圍，4-95 三條區間並列就看得很清楚。
- **Agresti-Coull**：只用在 4-89 一題，並且明確跟傳統 Wald CI 並列比較。

R 4.4.1 跑完所有題目，繪圖用 `ggplot2`；套件交叉驗證用 `BSDA::z.test`、`stats::t.test`、`stats::prop.test`、`tolerance::normtol.int`。

附錄 A：完整 R 原始碼 (homework1.R)

```
# ==============================================================================
# Homework Assignments 1  ----  Chapter 4 Single-Sample Inference
# Exercises: 4-42, 4-44, 4-54, 4-59, 4-65, 4-71, 4-75, 4-76, 4-89, 4-95
# ==============================================================================

## ---- init ----
# --- 1. Setup ------------------------------------------------------------
# 包管理用 pacman，看看哪幾顆要載
if (!require(pacman)) install.packages("pacman", repos = "https://cloud.r-project.org")
pacman::p_load(tidyverse, car, BSDA, tolerance, ggpubr)

# 圖會丟到 plots/，沒資料夾就建
dir.create("plots", showWarnings = FALSE)

# 簡單的標題分隔線，跟老師 console 風格一致
print_header <- function(title) {
  cat("\n====================================================\n")
  cat(" ", title, "\n", sep = "")
  cat("====================================================\n")
}

print_subhead <- function(title) {
  cat(sprintf("\n--- %s ---\n", title))
}

# --- 2. Shared Helper Functions ------------------------------------------

# 描述統計量，順便四捨五入
get_stats <- function(x) {
  data.frame(
    n         = length(x),
    Mean      = mean(x),
    Var       = var(x),
    Std_Dev   = sd(x),
    Std_Error = sd(x) / sqrt(length(x)),
    Median    = median(x)
  ) %>% mutate(across(everything(), ~round(., 4)))
}

# 已知 sigma 的 z 檢定 power
Pwr.z.test <- function(n, delta, sigma, alpha, alt = "two.sided") {
  m <- delta / (sigma / sqrt(n))
  if (alt == "two.sided") {
    1 - pnorm(qnorm(1 - alpha/2) - m) + pnorm(qnorm(alpha/2) - m)
  } else if (alt == "greater") {
    1 - pnorm(qnorm(1 - alpha) - m)
  } else {
    pnorm(qnorm(alpha) - m)
  }
}

# 已知 sigma 的 z 檢定樣本數
SplSz.z.test <- function(delta, sigma, alpha, pwr, side = 2) {
  ceiling(((qnorm(1 - alpha/side) + qnorm(pwr)) * sigma / abs(delta))^2)
}

# 已知 sigma 估平均數的 CI 樣本數 (margin of error 公式)
SplSz.z.CI <- function(E, sigma, conf_level, side = 2) {
```

```
  ceiling((qnorm(1 - (1 - conf_level)/side) * sigma / E)^2)
}

# 未知 sigma 的 t 檢定 power，用 noncentral t
Pwr.t.test.custom <- function(n, delta, sigma, alpha, alt = "two.sided") {
  df  <- n - 1
  ncp <- delta / (sigma / sqrt(n))
  if (alt == "two.sided") {
    1 - pt(qt(1 - alpha/2, df), df, ncp) + pt(qt(alpha/2, df), df, ncp)
  } else if (alt == "greater") {
    1 - pt(qt(1 - alpha, df), df, ncp)
  } else {
    pt(qt(alpha, df), df, ncp)
  }
}

# 迭代找最小 n，t 版本
find_min_n_t <- function(delta, sigma, alpha, target_power,
                         alt = "two.sided", n_start = 2, n_max = 5000) {
  for (n in n_start:n_max) {
    pw <- Pwr.t.test.custom(n, delta, sigma, alpha, alt)
    if (pw >= target_power) return(n)
  }
  stop(" 找不到符合條件的樣本數")
}

# 卡方檢定的 power (對 sigma 做檢定用)
Pwr.chisq.test <- function(n, r, alpha, alt = "two.sided") {
  df <- n - 1
  if (alt == "two.sided") {
    1 - pchisq(qchisq(1 - alpha/2, df) / r^2, df) +
      pchisq(qchisq(alpha/2, df) / r^2, df)
  } else if (alt == "greater") {
    1 - pchisq(qchisq(1 - alpha, df) / r^2, df)
  } else {
    pchisq(qchisq(alpha, df) / r^2, df)
  }
}

# 傳統 (Wald) 比例 CI
prop_wald_ci <- function(x, n, conf_level = 0.95, side = "two.sided") {
  phat  <- x / n
  alpha <- 1 - conf_level
  se    <- sqrt(phat * (1 - phat) / n)
  if (side == "two.sided") {
    z <- qnorm(1 - alpha/2)
    c(phat - z * se, phat + z * se)
  } else if (side == "lower") {
    z <- qnorm(conf_level)
    c(phat - z * se, Inf)
  } else {
    z <- qnorm(conf_level)
    c(-Inf, phat + z * se)
  }
}

# Agresti-Coull CI
agresti_coull_ci <- function(x, n, conf_level = 0.95) {
  alpha    <- 1 - conf_level
  z        <- qnorm(1 - alpha/2)
```

```
  n_tilde  <- n + z^2
  p_tilde  <- (x + z^2 / 2) / n_tilde
  se_tilde <- sqrt(p_tilde * (1 - p_tilde) / n_tilde)
  c(p_tilde - z * se_tilde, p_tilde + z * se_tilde)
}

# 一筆未來觀測的常態預測區間
prediction_interval_normal <- function(x, conf_level = 0.95) {
  n     <- length(x)
  xbar  <- mean(x)
  s     <- sd(x)
  alpha <- 1 - conf_level
  crit  <- qt(c(alpha/2, 1 - alpha/2), df = n - 1)
  xbar + crit * s * sqrt(1 + 1/n)
}

# Q-Q plot + 直方圖一起存，順便把 Shapiro-Wilk 結果與描述統計打到副標題
save_normality_plots <- function(x, title_tag, file_prefix) {
  df <- data.frame(value = x)
  sw <- shapiro.test(x)
  sw_sub <- sprintf("Shapiro-Wilk: W = %.4f, p = %.4f", sw$statistic, sw$p.value)
  stats_sub <- sprintf("n = %d, x_bar = %.4f, s = %.4f",
                       length(x), mean(x), sd(x))
  # Q-Q：點 + 理論直線 + Shapiro-Wilk 註解
  p_qq <- ggplot(df, aes(sample = value)) +
    stat_qq(color = "steelblue", size = 2.2) +
    stat_qq_line(color = "firebrick", linewidth = 0.8) +
    labs(title = paste("Normal Q-Q Plot:", title_tag),
         subtitle = sw_sub,
         x = "Theoretical Quantiles", y = "Sample Quantiles") +
    theme_minimal(base_size = 12)
  # 直方圖：n 太小 (<= 6) 用較少 bin，順便加 rug 把每筆觀測點都標出來
  bins_used <- if (length(x) <= 6) 4 else max(5, min(10, ceiling(log2(length(x)) + 1)))
  p_hist <- ggplot(df, aes(x = value)) +
    geom_histogram(aes(y = after_stat(density)),
                   bins = bins_used,
                   fill = "steelblue", color = "white", alpha = 0.85) +
    geom_rug(color = "navyblue", sides = "b", length = unit(0.04, "npc")) +
    stat_function(fun = dnorm,
                  args = list(mean = mean(x), sd = sd(x)),
                  color = "firebrick", linewidth = 1) +
    labs(title = paste("Histogram with Normal Curve:", title_tag),
         subtitle = stats_sub,
         x = "Value", y = "Density") +
    theme_minimal(base_size = 12)
  ggsave(paste0("plots/", file_prefix, "_qq.png"),
         p_qq,   width = 6.5, height = 4.5, dpi = 200)
  ggsave(paste0("plots/", file_prefix, "_hist.png"),
         p_hist, width = 6.5, height = 4.5, dpi = 200)
}

## ---- ex42 ----
# ==============================================================================
# EXERCISE 4-42: Thermocouple Life (Section 4-4)
# 已知 sigma=20, n=15, mu0=540, alpha=0.05, H1: mu > 540
# ==============================================================================
print_header("EXERCISE 4-42: THERMOCOUPLE LIFE")

# 原始資料，照題目 hard-code，不可改
thermo <- c(553, 552, 567, 579, 550, 541, 537, 553, 552,
```

```
            546, 538, 553, 581, 539, 529)

params_42 <- list(mu0 = 540, sigma = 20, alpha = 0.05, n = length(thermo))

print_subhead("Summary statistics")
print(get_stats(thermo))
cat(sprintf("Known sigma = %.4f\n", params_42$sigma))
cat(sprintf("Known SE    = %.4f\n", params_42$sigma / sqrt(params_42$n)))

# (a) z 檢定，one-sided greater
print_subhead("(a)(b) One-sided z-test, P-value approach")
(xbar     <- mean(thermo))
(se_42    <- params_42$sigma / sqrt(params_42$n))
(z0_42    <- (xbar - params_42$mu0) / se_42)
(p_42     <- 1 - pnorm(z0_42))
crit_42   <- qnorm(1 - params_42$alpha)

cat("H0: mu <= 540\n")
cat("H1: mu >  540\n")
cat(sprintf("alpha       = %.2f\n", params_42$alpha))
cat(sprintf("xbar        = %.4f\n", xbar))
cat(sprintf("z0          = %.4f\n", z0_42))
cat(sprintf("z critical  = %.4f\n", crit_42))
cat(sprintf("P-value     = %.4f\n", p_42))
cat("Decision: ",
    if (p_42 < params_42$alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")
cat("Conclusion: ",
    if (p_42 < params_42$alpha)
      "There is evidence that mean life exceeds 540 hours.\n"
    else
      "Insufficient evidence that mean life exceeds 540 hours.\n")

# 套件交叉驗證
zchk_42 <- BSDA::z.test(thermo, mu = params_42$mu0,
                        sigma.x = params_42$sigma,
                        alternative = "greater")
cat(sprintf("[Cross-check BSDA::z.test]  z = %.4f, p = %.4f\n",
            zchk_42$statistic, zchk_42$p.value))

# (c) Beta，當真實 mu = 560
print_subhead("(c) Beta when true mu = 560")
# 拒絕門檻換算成 xbar 比較直觀
xbar_crit_42 <- params_42$mu0 + crit_42 * se_42
mu_true_42   <- 560
(beta_42c    <- pnorm((xbar_crit_42 - mu_true_42) / se_42))
power_42c    <- 1 - beta_42c
cat(sprintf("xbar critical = %.4f\n", xbar_crit_42))
cat(sprintf("beta          = %.4f\n", beta_42c))
cat(sprintf("power         = %.4f\n", power_42c))

# (d) 要 beta <= 0.10 需要的 n
print_subhead("(d) Sample size so beta <= 0.10 when true mu = 560")
delta_42d <- mu_true_42 - params_42$mu0
n_req_42d <- SplSz.z.test(delta = delta_42d, sigma = params_42$sigma,
                          alpha = params_42$alpha, pwr = 0.90, side = 1)
pw_minus  <- Pwr.z.test(n_req_42d - 1, delta_42d, params_42$sigma,
                        params_42$alpha, alt = "greater")
pw_at     <- Pwr.z.test(n_req_42d,     delta_42d, params_42$sigma,
                        params_42$alpha, alt = "greater")
cat(sprintf("Required n         = %d\n", n_req_42d))
```

```
cat(sprintf("Power at n-1 (%2d) = %.4f , beta = %.4f\n",
            n_req_42d - 1, pw_minus, 1 - pw_minus))
cat(sprintf("Power at n   (%2d) = %.4f , beta = %.4f\n",
            n_req_42d,     pw_at,    1 - pw_at))

# (e) 95% one-sided lower CI on mu
print_subhead("(e) 95% one-sided lower CI on mean life")
LCL_42 <- xbar - crit_42 * se_42
cat(sprintf("LCL = %.4f\n", LCL_42))
cat(sprintf("CI : [ %.4f , Inf )\n", LCL_42))

# (f) 用 CI 做檢定
print_subhead("(f) Test the hypothesis using the CI in (e)")
cat(sprintf("LCL = %.4f, mu0 = %d => ", LCL_42, params_42$mu0))
if (LCL_42 > params_42$mu0) {
  cat("LCL > 540, the null value lies outside the CI, reject H0.\n")
} else {
  cat("LCL <= 540, the null value lies inside the CI, do not reject H0.\n")
}

# Power curve 視覺化
power_grid_42 <- tibble(
  mu_true = seq(540, 580, length.out = 200)
) %>% mutate(
  power = Pwr.z.test(params_42$n, mu_true - params_42$mu0,
                     params_42$sigma, params_42$alpha, alt = "greater")
)
p_pow_42 <- ggplot(power_grid_42, aes(mu_true, power)) +
  geom_line(color = "steelblue", linewidth = 1) +
  geom_hline(yintercept = 0.90, linetype = "dashed", color = "firebrick") +
  geom_vline(xintercept = 560, linetype = "dotted",  color = "darkgreen") +
  labs(title = "Ex 4-42 Power curve",
       subtitle = "known   = 20, n = 15,   = 0.05",
       x = "True mean life", y = "Power") +
  theme_minimal(base_size = 12)
ggsave("plots/ex4_42_power_curve.png", p_pow_42,
       width = 6.5, height = 4.5, dpi = 200)


## ---- ex44 ----
# ==============================================================================
# EXERCISE 4-44: Sample size for estimating the mean (Section 4-4)
# 沿用 4-42 的 sigma = 20，要 95% 信賴、誤差 < 5 小時
# ==============================================================================
print_header("EXERCISE 4-44: SAMPLE SIZE FOR MEAN LIFE")

E_44       <- 5
sigma_44   <- 20
conf_44    <- 0.95
n_44       <- SplSz.z.CI(E = E_44, sigma = sigma_44,
                         conf_level = conf_44, side = 2)

cat(sprintf("Confidence level   = %.2f\n", conf_44))
cat(sprintf("Margin of error E  = %.0f\n", E_44))
cat(sprintf("sigma              = %.0f\n", sigma_44))
cat(sprintf("Required sample n  = %d\n", n_44))
# 用公式手算對一下
z_44 <- qnorm(1 - (1 - conf_44)/2)
cat(sprintf("Manual: ((z * sigma)/E)^2 = ((%.4f * %d)/%d)^2 = %.4f -> ceiling = %d\n",
            z_44, sigma_44, E_44, (z_44 * sigma_44 / E_44)^2, n_44))
```

```r


## ---- ex54 ----
# ==============================================================================
# EXERCISE 4-54: Interior Temperature  (Section 4-5)
# 未知 sigma, n=5, mu0=22.5, alpha=0.05, H1: mu != 22.5
# ==============================================================================
print_header("EXERCISE 4-54: INTERIOR TEMPERATURE")

temp_int <- c(23.01, 22.22, 22.04, 22.62, 22.59)
params_54 <- list(mu0 = 22.5, alpha = 0.05, n = length(temp_int))

print_subhead("Summary statistics")
print(get_stats(temp_int))

# (b) 常態性檢查
print_subhead("(b) Normality check  (Shapiro-Wilk + Q-Q plot)")
(sw_54 <- shapiro.test(temp_int))
cat(" 樣本數很小，Shapiro-Wilk power 不高，需要搭配 Q-Q plot 一起看。\n")
save_normality_plots(temp_int, "Ex 4-54 Interior Temperature", "ex4_54")
cat("Plots saved: plots/ex4_54_qq.png, plots/ex4_54_hist.png\n")

# (a) t 檢定，雙尾
print_subhead("(a) Two-sided t-test")
t_54 <- t.test(temp_int, mu = params_54$mu0, alternative = "two.sided",
               conf.level = 0.95)
xbar_54 <- mean(temp_int)
s_54    <- sd(temp_int)
se_54   <- s_54 / sqrt(params_54$n)
t0_54   <- (xbar_54 - params_54$mu0) / se_54
p_54    <- 2 * (1 - pt(abs(t0_54), df = params_54$n - 1))

cat("H0: mu  = 22.5\n")
cat("H1: mu != 22.5\n")
cat(sprintf("xbar     = %.4f\n", xbar_54))
cat(sprintf("s        = %.4f\n", s_54))
cat(sprintf("t0       = %.4f\n", t0_54))
cat(sprintf("df       = %d\n",   params_54$n - 1))
cat(sprintf("P-value  = %.4f\n", p_54))
cat("Decision: ",
    if (p_54 < params_54$alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")
cat(sprintf("[t.test cross-check]  t = %.4f, p = %.4f\n",
            t_54$statistic, t_54$p.value))

# (c) 95% CI
print_subhead("(c) 95% CI for mean interior temperature")
CI_54 <- as.numeric(t_54$conf.int)
cat(sprintf("CI = (%.4f , %.4f)\n", CI_54[1], CI_54[2]))

# (d) 樣本數，要 power >= 0.9 偵測 mu=22.75
print_subhead("(d) Sample size for power >= 0.9, true mu = 22.75")
delta_54d  <- 22.75 - params_54$mu0
sigma_54d  <- s_54
n_req_54d  <- find_min_n_t(delta = delta_54d, sigma = sigma_54d,
                           alpha = params_54$alpha,
                           target_power = 0.90, alt = "two.sided")
pw_minus54 <- Pwr.t.test.custom(n_req_54d - 1, delta_54d, sigma_54d,
                                params_54$alpha, alt = "two.sided")
pw_at_54   <- Pwr.t.test.custom(n_req_54d,     delta_54d, sigma_54d,
                                params_54$alpha, alt = "two.sided")
```

```
cat(sprintf("delta       = %.4f, sigma (=s) = %.4f\n", delta_54d, sigma_54d))
cat(sprintf("Required n  = %d\n", n_req_54d))
cat(sprintf("Power at n-1 (%d) = %.4f\n", n_req_54d - 1, pw_minus54))
cat(sprintf("Power at n   (%d) = %.4f\n", n_req_54d,     pw_at_54))


## ---- ex59 ----
# ==============================================================================
# EXERCISE 4-59: Diode Breakdown Voltage (Section 4-5)
# n=12, mu0=9, alpha=0.05, H1: mu < 9
# ==============================================================================
print_header("EXERCISE 4-59: DIODE BREAKDOWN VOLTAGE")

diode_v <- c(9.099, 9.174, 9.327, 9.377, 8.471, 9.575,
             9.514, 8.928, 8.800, 8.920, 9.913, 8.306)

params_59 <- list(mu0 = 9, alpha = 0.05, n = length(diode_v))

print_subhead("Summary statistics")
print(get_stats(diode_v))

# (a) 常態性
print_subhead("(a) Normality check")
(sw_59 <- shapiro.test(diode_v))
save_normality_plots(diode_v, "Ex 4-59 Diode Voltage", "ex4_59")
cat("Plots saved: plots/ex4_59_qq.png, plots/ex4_59_hist.png\n")

# (b) t 檢定，one-sided less
print_subhead("(b) One-sided t-test, H1: mu < 9")
t_59      <- t.test(diode_v, mu = params_59$mu0, alternative = "less",
                    conf.level = 0.95)
xbar_59   <- mean(diode_v)
s_59      <- sd(diode_v)
se_59     <- s_59 / sqrt(params_59$n)
t0_59     <- (xbar_59 - params_59$mu0) / se_59
p_59      <- pt(t0_59, df = params_59$n - 1)

cat(sprintf("xbar    = %.4f\n", xbar_59))
cat(sprintf("s       = %.4f\n", s_59))
cat(sprintf("t0      = %.4f\n", t0_59))
cat(sprintf("df      = %d\n",   params_59$n - 1))
cat(sprintf("P-value = %.4f\n", p_59))
cat("Decision: ",
    if (p_59 < params_59$alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")
cat(sprintf("[t.test cross-check]  t = %.4f, p = %.4f\n",
            t_59$statistic, t_59$p.value))

# (c) 95% one-sided upper bound
print_subhead("(c) 95% one-sided upper confidence bound")
UCL_59 <- xbar_59 + qt(1 - params_59$alpha, df = params_59$n - 1) * se_59
cat(sprintf("UCL = %.4f\n", UCL_59))
cat(sprintf("CI : ( -Inf , %.4f ]\n", UCL_59))

# (d) 用 UCL 做檢定
print_subhead("(d) Test using the upper bound from (c)")
cat(sprintf("UCL = %.4f, mu0 = %d => ", UCL_59, params_59$mu0))
if (UCL_59 < params_59$mu0) {
  cat("UCL < 9, mu0 lies outside the CI, reject H0.\n")
} else {
  cat("UCL >= 9, mu0 lies inside the CI, do not reject H0.\n")
```

```
}

# (e) 樣本數要 power >= 0.95, 偵測 mu=8.8
print_subhead("(e) Sample size for power >= 0.95, true mu = 8.8")
delta_59e <- 8.8 - params_59$mu0
sigma_59e <- s_59
n_req_59e <- find_min_n_t(delta = delta_59e, sigma = sigma_59e,
                          alpha = params_59$alpha,
                          target_power = 0.95, alt = "less")
pw_minus59 <- Pwr.t.test.custom(n_req_59e - 1, delta_59e, sigma_59e,
                                params_59$alpha, alt = "less")
pw_at_59   <- Pwr.t.test.custom(n_req_59e,     delta_59e, sigma_59e,
                                params_59$alpha, alt = "less")
cat(sprintf("delta = %.4f, sigma (=s) = %.4f\n", delta_59e, sigma_59e))
cat(sprintf("Required n = %d\n", n_req_59e))
cat(sprintf("Power at n-1 (%d) = %.4f\n", n_req_59e - 1, pw_minus59))
cat(sprintf("Power at n   (%d) = %.4f\n", n_req_59e,     pw_at_59))


## ---- ex65 ----
# ==============================================================================
# EXERCISE 4-65: Cloud Seeding Rainfall (Section 4-5)
# n=20, mu0=25, alpha=0.01, H1: mu > 25
# Note: 題目 (e) 寫 mean diameter，但全題討論的是 rainfall，疑似誤植
# ==============================================================================
print_header("EXERCISE 4-65: CLOUD SEEDING RAINFALL")

rain <- c(18.0, 30.7, 19.8, 27.1, 22.3, 18.8, 31.8, 23.4, 21.2, 27.9,
          31.9, 27.1, 25.0, 24.7, 26.9, 21.8, 29.2, 34.8, 26.7, 31.6)

params_65 <- list(mu0 = 25, alpha = 0.01, n = length(rain))

print_subhead("Summary statistics")
print(get_stats(rain))

# (b) 常態性 (先看圖再做 t test)
print_subhead("(b) Normality check")
(sw_65 <- shapiro.test(rain))
save_normality_plots(rain, "Ex 4-65 Cloud Seeding Rainfall", "ex4_65")
cat("Plots saved: plots/ex4_65_qq.png, plots/ex4_65_hist.png\n")

# (a) t 檢定，one-sided greater
print_subhead("(a) One-sided t-test, P-value")
t_65    <- t.test(rain, mu = params_65$mu0, alternative = "greater",
                  conf.level = 0.99)
xbar_65 <- mean(rain)
s_65    <- sd(rain)
se_65   <- s_65 / sqrt(params_65$n)
t0_65   <- (xbar_65 - params_65$mu0) / se_65
p_65    <- 1 - pt(t0_65, df = params_65$n - 1)
cat(sprintf("xbar     = %.4f\n", xbar_65))
cat(sprintf("s        = %.4f\n", s_65))
cat(sprintf("t0       = %.4f\n", t0_65))
cat(sprintf("df       = %d\n",   params_65$n - 1))
cat(sprintf("P-value  = %.4f\n", p_65))
cat("Decision: ",
    if (p_65 < params_65$alpha) "REJECT H0" else "FAIL TO REJECT H0", "\n")
cat(sprintf("[t.test cross-check]  t = %.4f, p = %.4f\n",
            t_65$statistic, t_65$p.value))

```

```
# (c) Power, true mu = 27
print_subhead("(c) Power if true mu = 27")
delta_65c <- 27 - params_65$mu0
pw_65c    <- Pwr.t.test.custom(params_65$n, delta_65c, s_65,
                               params_65$alpha, alt = "greater")
cat(sprintf("delta = %.4f, sigma (=s) = %.4f\n", delta_65c, s_65))
cat(sprintf("Power = %.4f, beta = %.4f\n", pw_65c, 1 - pw_65c))

# (d) 樣本數，要 power >= 0.9，true mu = 27.5
print_subhead("(d) Sample size for power >= 0.9, true mu = 27.5")
delta_65d  <- 27.5 - params_65$mu0
n_req_65d  <- find_min_n_t(delta = delta_65d, sigma = s_65,
                           alpha = params_65$alpha,
                           target_power = 0.90, alt = "greater")
pw_minus65 <- Pwr.t.test.custom(n_req_65d - 1, delta_65d, s_65,
                               params_65$alpha, alt = "greater")
pw_at_65   <- Pwr.t.test.custom(n_req_65d,     delta_65d, s_65,
                               params_65$alpha, alt = "greater")
cat(sprintf("Required n = %d\n", n_req_65d))
cat(sprintf("Power at n-1 (%d) = %.4f\n", n_req_65d - 1, pw_minus65))
cat(sprintf("Power at n   (%d) = %.4f\n", n_req_65d,     pw_at_65))

# (e) 一邊下界 CI 解釋
# 題目寫 mean diameter，但脈絡是 rainfall，這邊照題目敘述但加註
print_subhead("(e) One-sided lower confidence bound for mean rainfall")
cat("Note: The textbook says 'mean diameter' in part (e); the context indicates ")
cat("this should be 'mean rainfall'. We answer for mean rainfall.\n")
LCL_65e <- xbar_65 - qt(1 - params_65$alpha, df = params_65$n - 1) * se_65
cat(sprintf("LCL = %.4f\n", LCL_65e))
cat(sprintf("CI : [ %.4f , Inf )\n", LCL_65e))
cat(sprintf("Compare LCL with mu0 = %d: ", params_65$mu0))
if (LCL_65e > params_65$mu0) {
  cat("LCL > 25, reject H0 -- supports the claim.\n")
} else {
  cat("LCL <= 25, do not reject H0.\n")
}


## ---- ex71 ----
# ==============================================================================
# EXERCISE 4-71: Titanium Alloy Standard Deviation (Section 4-6)
# n=51, s=0.37, sigma0=0.35, alpha=0.05, two-sided
# ==============================================================================
print_header("EXERCISE 4-71: TITANIUM ALLOY SIGMA TEST")

# 題目只給 summary，沒有原始資料，hard-code 即可
n_71      <- 51
s_71      <- 0.37
sigma0_71 <- 0.35
alpha_71  <- 0.05
df_71     <- n_71 - 1

cat(" 假設母體近似常態分配，這個 chi-square test 才有效。\n")

# (a)(b) 卡方統計量、P-value
print_subhead("(a)(b) Chi-square test")
chi0_71   <- (n_71 - 1) * s_71^2 / sigma0_71^2
# 雙尾 P-value 取 2 * min(left, right)
p_left    <- pchisq(chi0_71, df_71)
p_right   <- 1 - pchisq(chi0_71, df_71)
```

```
p_71      <- 2 * min(p_left, p_right)

cat("H0: sigma  = 0.35\n")
cat("H1: sigma != 0.35\n")
cat(sprintf("chi0    = %.4f\n", chi0_71))
cat(sprintf("df      = %d\n",   df_71))
cat(sprintf("P-value = %.4f\n", p_71))
cat(sprintf("Critical: chi^2_{0.025,%d} = %.4f, chi^2_{0.975,%d} = %.4f\n",
            df_71, qchisq(0.975, df_71), df_71, qchisq(0.025, df_71)))
cat("Decision: ",
    if (p_71 < alpha_71) "REJECT H0" else "FAIL TO REJECT H0", "\n")

# (c) 95% CI for sigma
# 先做 sigma^2 的 CI 再開根號
print_subhead("(c) 95% two-sided CI for sigma")
LCL_var <- (n_71 - 1) * s_71^2 / qchisq(1 - alpha_71/2, df_71)
UCL_var <- (n_71 - 1) * s_71^2 / qchisq(    alpha_71/2, df_71)
LCL_sd  <- sqrt(LCL_var)
UCL_sd  <- sqrt(UCL_var)
cat(sprintf("CI for sigma^2 : (%.4f , %.4f)\n", LCL_var, UCL_var))
cat(sprintf("CI for sigma   : (%.4f , %.4f)\n", LCL_sd,  UCL_sd))

# (d) 用 CI 做檢定
print_subhead("(d) Test the hypothesis using the CI")
cat(sprintf("Is sigma0 = %.2f inside (%.4f, %.4f)? ", sigma0_71, LCL_sd, UCL_sd))
if (sigma0_71 >= LCL_sd && sigma0_71 <= UCL_sd) {
  cat("Yes => do not reject H0.\n")
} else {
  cat("No  => reject H0.\n")
}


## ---- ex75 ----
# ==============================================================================
# EXERCISE 4-75: Van Rollover Proportion (Section 4-7)
# n=30, x=11, p0=0.25, alpha=0.10, H1: p > 0.25
# ==============================================================================
print_header("EXERCISE 4-75: VAN ROLLOVER PROPORTION")

n_75    <- 30
x_75    <- 11
p0_75   <- 0.25
alpha_75 <- 0.10
phat_75 <- x_75 / n_75

cat(sprintf("phat = %d / %d = %.4f\n", x_75, n_75, phat_75))

# (a) z 檢定
print_subhead("(a) One-proportion z-test, H1: p > 0.25")
se0_75 <- sqrt(p0_75 * (1 - p0_75) / n_75)
z0_75  <- (phat_75 - p0_75) / se0_75
p_75   <- 1 - pnorm(z0_75)
cat(sprintf("z0      = %.4f\n", z0_75))
cat(sprintf("P-value = %.4f\n", p_75))
cat("Decision: ",
    if (p_75 < alpha_75) "REJECT H0" else "FAIL TO REJECT H0", "\n")
# 套件交叉驗證 (correct=FALSE 不做連續性修正)
ptest_75 <- prop.test(x_75, n_75, p = p0_75, alternative = "greater",
                      correct = FALSE)
cat(sprintf("[prop.test cross-check]  chi^2 = %.4f, p = %.4f\n",
```

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            ptest_75$statistic, ptest_75$p.value))

# (b) Beta，true p = 0.35
print_subhead("(b) Beta when true p = 0.35")
# 在 H0 下找臨界 phat
phat_crit <- p0_75 + qnorm(1 - alpha_75) * se0_75
true_p    <- 0.35
se_true   <- sqrt(true_p * (1 - true_p) / n_75)
beta_75b  <- pnorm((phat_crit - true_p) / se_true)
cat(sprintf("phat critical = %.4f\n", phat_crit))
cat(sprintf("beta          = %.4f\n", beta_75b))

# (c) 樣本數要 beta = 0.10
print_subhead("(c) Sample size so beta = 0.10 when true p = 0.35")
z_alpha <- qnorm(1 - alpha_75)
z_beta  <- qnorm(1 - 0.10)
num_75  <- z_alpha * sqrt(p0_75*(1-p0_75)) + z_beta * sqrt(true_p*(1-true_p))
n_75c   <- ceiling((num_75 / (true_p - p0_75))^2)
cat(sprintf("Required n = %d\n", n_75c))

# (d) 90% one-sided lower bound
print_subhead("(d) 90% traditional one-sided lower confidence bound")
LCB_75 <- prop_wald_ci(x_75, n_75, conf_level = 0.90, side = "lower")[1]
cat(sprintf("LCB = %.4f\n", LCB_75))
cat(sprintf("CI : [ %.4f , 1 ]\n", LCB_75))

# (e) 用 LCB 做檢定
print_subhead("(e) Test using the lower bound")
cat(sprintf("LCB = %.4f vs p0 = %.2f => ", LCB_75, p0_75))
if (LCB_75 > p0_75) {
  cat("LCB > 0.25, reject H0 -- claim supported.\n")
} else {
  cat("LCB <= 0.25, do not reject H0.\n")
}

# (f) 樣本數要誤差 < 0.02，95% confident，用 phat = 11/30
print_subhead("(f) Sample size for E < 0.02, 95% confidence, phat = 11/30")
z_95   <- qnorm(0.975)
n_75f  <- ceiling(z_95^2 * phat_75 * (1 - phat_75) / 0.02^2)
cat(sprintf("Using phat = %.4f, required n = %d\n", phat_75, n_75f))


## ---- ex76 ----
# ==============================================================================
# EXERCISE 4-76: Helmet Damage Proportion (Section 4-7)
# n=50, x=18, p0=0.3, alpha=0.05, H1: p != 0.3
# ==============================================================================
print_header("EXERCISE 4-76: HELMET DAMAGE PROPORTION")

n_76    <- 50
x_76    <- 18
p0_76   <- 0.30
alpha_76 <- 0.05
phat_76 <- x_76 / n_76

cat(sprintf("phat = %d / %d = %.4f\n", x_76, n_76, phat_76))

# (a)(b) z 檢定 + P-value
print_subhead("(a)(b) Two-sided z-test")
se0_76 <- sqrt(p0_76 * (1 - p0_76) / n_76)
```

```
z0_76  <- (phat_76 - p0_76) / se0_76
p_76   <- 2 * (1 - pnorm(abs(z0_76)))
cat(sprintf("z0      = %.4f\n", z0_76))
cat(sprintf("P-value = %.4f\n", p_76))
cat("Decision: ",
    if (p_76 < alpha_76) "REJECT H0" else "FAIL TO REJECT H0", "\n")
# 套件交叉驗證
ptest_76 <- prop.test(x_76, n_76, p = p0_76, alternative = "two.sided",
                      correct = FALSE)
cat(sprintf("[prop.test cross-check]  chi^2 = %.4f, p = %.4f\n",
            ptest_76$statistic, ptest_76$p.value))

# (c) 95% two-sided traditional CI
print_subhead("(c) 95% two-sided traditional CI")
CI_76 <- prop_wald_ci(x_76, n_76, conf_level = 0.95, side = "two.sided")
cat(sprintf("CI = (%.4f , %.4f)\n", CI_76[1], CI_76[2]))
cat(sprintf("Is p0 = %.2f inside the CI? ", p0_76))
if (p0_76 >= CI_76[1] && p0_76 <= CI_76[2]) {
  cat("Yes => fail to reject H0 (consistent with the test).\n")
} else {
  cat("No  => reject H0.\n")
}

# (d) 樣本數要 E < 0.02，用 phat = 18/50
print_subhead("(d) Sample size for E < 0.02, 95% conf., using phat = 18/50")
z95_76 <- qnorm(0.975)
n_76d  <- ceiling(z95_76^2 * phat_76 * (1 - phat_76) / 0.02^2)
cat(sprintf("Using phat = %.4f, required n = %d\n", phat_76, n_76d))

# (e) 保守樣本數，用 p = 0.5
print_subhead("(e) Conservative sample size using p = 0.5")
n_76e <- ceiling(z95_76^2 * 0.5 * 0.5 / 0.02^2)
cat(sprintf("Using p = 0.50, required n = %d\n", n_76e))


## ---- ex89 ----
# ==============================================================================
# EXERCISE 4-89: Agresti-Coull CI vs Traditional (Section 4-7)
# 用 4-76 的資料: n=50, x=18, 95% confidence
# ==============================================================================
print_header("EXERCISE 4-89: AGRESTI-COULL CI")

ac_89 <- agresti_coull_ci(x_76, n_76, conf_level = 0.95)

# Agresti-Coull 是把 (x, n) 換成 (x + z^2/2, n + z^2)
z_89   <- qnorm(0.975)
n_t_89 <- n_76 + z_89^2
p_t_89 <- (x_76 + z_89^2 / 2) / n_t_89

cat("Agresti-Coull 用調整過的樣本數和點估計來建立 CI，這是和傳統 Wald CI 的核心差別。\n")
cat(sprintf("Adjusted n_tilde = %.4f\n", n_t_89))
cat(sprintf("Adjusted p_tilde = %.4f\n", p_t_89))
cat(sprintf("Agresti-Coull CI : (%.4f , %.4f)\n", ac_89[1], ac_89[2]))
cat(sprintf("Traditional CI   : (%.4f , %.4f)\n", CI_76[1], CI_76[2]))
cat(sprintf("Width comparison : AC width = %.4f, traditional width = %.4f\n",
            ac_89[2] - ac_89[1], CI_76[2] - CI_76[1]))


## ---- ex95 ----
# ==============================================================================
```

```r
# EXERCISE 4-95: Prediction & Tolerance Interval (Section 4-8)
# 沿用 4-59 的二極體資料，99% PI 與 99% / 99% TI
# ==============================================================================
print_header("EXERCISE 4-95: PI AND TOLERANCE INTERVAL")

# (a) 99% PI for one future observation
print_subhead("(a) 99% Prediction Interval for one future diode")
PI_95 <- prediction_interval_normal(diode_v, conf_level = 0.99)
cat(sprintf("xbar = %.4f, s = %.4f, n = %d\n",
            mean(diode_v), sd(diode_v), length(diode_v)))
cat(sprintf("PI = (%.4f , %.4f)\n", PI_95[1], PI_95[2]))
cat("PI 的標的是『下一筆觀測值』，比 CI 寬一些是正常的。\n")

# (b) Tolerance interval, 99% population coverage with 99% confidence
print_subhead("(b) 99%/99% two-sided normal tolerance interval")
TI_95 <- tolerance::normtol.int(diode_v, alpha = 0.01, P = 0.99, side = 2)
print(TI_95)
cat(sprintf("\nTI = (%.4f , %.4f)\n",
            TI_95$`2-sided.lower`, TI_95$`2-sided.upper`))
cat("TI 是希望涵蓋『母體中 99% 的個體』，而不是平均數，所以會比 PI 再寬一點。\n")

# 視覺化 PI / TI / CI 三者
ci_95 <- t.test(diode_v, conf.level = 0.99)$conf.int
intervals_df <- tibble(
  type   = c("99% CI for mean", "99% PI", "99% / 99% TI"),
  lower  = c(ci_95[1], PI_95[1], TI_95$`2-sided.lower`),
  upper  = c(ci_95[2], PI_95[2], TI_95$`2-sided.upper`)
) %>% mutate(width = upper - lower)
print(intervals_df)

p_intvl <- ggplot(intervals_df, aes(y = fct_inorder(type))) +
  geom_segment(aes(x = lower, xend = upper, yend = type),
               linewidth = 1.5, color = "steelblue") +
  geom_point(aes(x = lower), size = 3, color = "firebrick") +
  geom_point(aes(x = upper), size = 3, color = "firebrick") +
  geom_vline(xintercept = mean(diode_v), linetype = "dashed",
             color = "darkgreen") +
  labs(title = "Ex 4-95: CI, PI, and Tolerance Interval (diode voltage)",
       x = "Breakdown voltage", y = NULL) +
  theme_minimal(base_size = 12)
ggsave("plots/ex4_95_intervals.png", p_intvl,
       width = 7, height = 4, dpi = 200)
cat("Plot saved: plots/ex4_95_intervals.png\n")


# ==============================================================================
# 結束
# ==============================================================================
cat("\n====================================================\n")
cat(" All exercises completed.\n")
cat("====================================================\n")
```